CHRIST (Deemed to be University)

School of Business and Management

Pune, Lavasa Campus

Master of Business Administration (MBA)

Batch 2025-27

Trimester IV

Financial and Risk Analytics in the Insurance Sector

Financial Statements, Market Risk, Claim-Risk Prediction and Anomaly Screening

Submitted by	Alex Abraham - 25121007
Class	4MBA BA PLC
Course Code & Name	FINANCIAL AND RISK ANALYTICS - MBA441B
Assessment	CIA-4: Sector Analysis (Insurance and Risk Analytics)
Campus of Study	Pune Lavasa Campus
Faculty In-charge	Dr. ABHIJEET BIRARI
Date	9-10-2026

Abstract

This report applies financial and risk analytics to the Indian insurance sector using four separate pieces of evidence. The first is the financial statements of ICICI Lombard General Insurance for FY2024 to FY2026, taken from a workbook that was checked line by line against the audited annual reports and corrected where needed. Gross Direct Premium Income rose from ₹24,776 crore to ₹28,712 crore and profit after tax from ₹1,918.59 crore to ₹2,771.94 crore. The combined ratio stayed slightly above 100% (103.3%, 102.8% and 103.4%), so profit came from investment income rather than from underwriting. Solvency remained between 2.62 and 2.69 times against a regulatory minimum of 1.50. The second is 104 monthly returns of ICICI Lombard, GIC Re and SBI Life against the NIFTY 50. SBI Life and ICICI Lombard earned Sharpe ratios of 0.340 and 0.330, above the index (0.291), while GIC Re had 44.35% annualised volatility and a negative compound return. A long-only maximum-Sharpe portfolio of ICICI Lombard and SBI Life reached a Sharpe ratio of 0.372. The third is a claim-occurrence model built on the French freMTPL2freq motor dataset (678,013 policies, 5.02% with a claim). A gradient boosting model selected on validation data reached a test ROC-AUC of 0.665 and an Average Precision of 0.126, compared with 0.639 and 0.104 for the earlier random forest. Its riskiest 5% of policies had an observed claim rate of 17.6%. A separate Poisson frequency model that uses exposure explained 8.2% of deviance on the test set. The fourth is a pair of Isolation Forest screens that flagged about 1% of records each for review. None of these flags is evidence of fraud, and none of the models has been validated on Indian data.

Index

Section	Title	Page
	Abstract	2
	List of Abbreviations and Symbols	7
1	Introduction	8
1.1	Background and Motivation	8
1.2	Scope and Structure of the Report	8
2	Insurance Sector Overview	9
2.1	Structure and Business Model	9
2.2	Revenue, Costs and Capital	9
2.3	Major Risks	9
2.4	Indian Context and the Role of Analytics	10
3	Research Objectives	11
3.1	Objectives	11
4	Literature Review and Theoretical Framework	12
4.1	Financial Statement and Insurance Ratio Analysis	12
4.2	Risk, Return and Portfolio Theory	12
4.3	Claim-Risk Modelling and Classification	12
4.4	Anomaly Detection and Decision Support	12
4.5	AI, Generative AI and Responsible Analytics	13
4.6	Theoretical Framework	13
5	Data Sources and Preparation	14
5.1	Dataset Inventory	14
5.2	Financial Statement Data and Workbook Validation	14
5.3	Monthly Stock Data	14
5.4	The freMTPL2freq Claim Data	15
6	Financial Statement and Insurance Ratio Analysis	16
6.1	Growth and Profitability	16
6.2	Horizontal and Vertical Analysis	17
6.3	Underwriting Ratios and Solvency	18
6.4	Balance Sheet and Cash Flow	19
6.5	Piotroski-Style Score	20
6.6	Implications	21
7	Stock Market Risk and Portfolio Analysis	22
7.1	Method	22
7.2	Return and Risk of the Individual Stocks	22
7.3	Correlation, Beta and CAPM	24
7.4	Portfolio Construction and Performance	25
7.5	Implications	26
8	Insurance Claim-Risk Prediction Framework	27
8.1	Problem Definition and Target	27
8.2	Data Inspection	27
8.3	Preprocessing, Partitioning and Models	27
8.4	Model Comparison and Threshold Selection	28
8.5	Test Evaluation	29
8.6	Calibration, Risk Bands and Stability	30
8.7	Feature Importance and Partial Dependence	31
8.8	Claim-Frequency Model with Exposure	31

Section	Title	Page
8.9	Limitations and Possible Use	32
9	Anomaly Detection	33
9.1	Purpose and Method	33
9.2	Two Screens	33
9.3	Results	33
9.4	Interpretation and Leakage	34
9.5	Operational Use	34
10	Prediction Dashboard and Decision Support	35
10.1	Purpose and Workflow	35
10.2	Implemented Functionality	35
10.3	Illustrative Output	37
10.4	Human Oversight and Limitations	38
11	Role of Data Analytics, AI and Generative AI in Insurance	39
11.1	Types of Analytics	39
11.2	Applications and the Place of Generative AI	39
12	Integrated Discussion and Business Implications	40
12.1	Financial Performance, Risk and Investment	40
12.2	Predictive Modelling, Anomalies and Governance	40
13	Risk Mitigation and Recommendations	42
14	Limitations and Conclusion	43
14.1	Limitations	43
14.3	Conclusion	44
	References	45
	Appendix A: Supplementary Model Tables	48
	Appendix B: Reproducibility Notes	49

List of Figures

Figure No.	Figure Title	Page
Figure 1	GDPI and PAT of ICICI Lombard (₹ crore)	16
Figure 2	Segment operating profit and composition of Total Income (A)	17
Figure 3	Combined ratio and solvency ratio	18
Figure 4	Insurance operating ratios, FY2024 to FY2026	18
Figure 5	Cash flow by activity (₹ crore)	20
Figure 6	Growth of ₹1 invested in October 2017	23
Figure 7	Full-period and rolling 12-month annualised volatility	23
Figure 8	Historical risk and return	24
Figure 9	Correlation matrix of monthly returns	24
Figure 10	Portfolio return, risk and Sharpe ratio	25
Figure 11	Opportunity set of 10,000 random portfolios with the efficient boundary	26
Figure 12	Observed claim rate by predictor, all 678,013 policies	27
Figure 13	ROC curve, precision and recall, and cumulative gains on the test set	29
Figure 14	Calibration by decile and observed claim rate by risk band (test)	30
Figure 15	Permutation importance and partial dependence of the selected model	31
Figure 16	Calibration of the frequency model by decile (test)	32
Figure 17	Anomaly scores of both screens and screen B flags by claims and exposure	33
Figure 18	Workflow of the enhanced claim-risk framework	35
Figure 19	Policy Claim Probability Predictor with its input controls and result panel (driver aged 22, Bonus-Malus 90, one-year-old vehicle)	37

List of Tables

Table No.	Table Title	Page
Table 1	Abbreviations and symbols	7
Table 2	Analytical components, evidence and chapters	8
Table 3	Main insurer risks and where they are addressed	9
Table 4	Research questions and methods	11
Table 5	Theory, analysis and evidence	13
Table 6	Datasets used	14
Table 7	Main corrections made to the CIA-4 workbook	14
Table 8	Headline indicators, FY2024 to FY2026	16
Table 9	Horizontal analysis of selected P&L lines (₹ crore)	17
Table 10	Insurance operating ratios and solvency	18
Table 11	Balance-sheet and cash-flow summary (₹ crore)	19
Table 12	Piotroski criteria after correction	20
Table 13	Return and risk of the insurance stocks and NIFTY 50	22
Table 14	Beta and CAPM required return	25
Table 15	Portfolio weights and performance (annualised, R_f = 6%)	25
Table 16	Validation results of the candidate models (n = 135,603)	28
Table 17	Alert-threshold options on the validation set	28
Table 18	Test-set results (n = 135,603, scored once)	29
Table 19	Claims captured at different review capacities (test)	30
Table 20	Poisson frequency models (claims capped at 4, exposure at 1 year)	31
Table 21	Configuration of the two anomaly screens	33
Table 22	Summary of the two screens	33
Table 23	Functions of the predictor and notebook	35
Table 24	Analytics approaches and their use in this project	39
Table 25	Integrated findings	40
Table 26	Prioritised recommendations	42
Table 27	Lift by decile of predicted risk (test)	48
Table 28	Model stability by geographic area (test)	48
Table 29	Files, settings and software	49

List of Abbreviations and Symbols

Table 1: Abbreviations and symbols

Term	Meaning
AP	Average Precision (area under the precision and recall curve)
AUC	Area under the receiver operating characteristic (ROC) curve
β (beta)	Systematic risk: $Cov(R_i, R_m) / Var(R_m)$
BM	Bonus-Malus score (French no-claims discount and surcharge scale)
CAGR	Compound annual growth rate
CAPM	Capital Asset Pricing Model
CFO	Cash flow from operating activities
D^2	Share of Poisson deviance explained relative to a constant-frequency model
EOM	Expenses of management
FVCA	Fair value change account
FY	Financial year ending 31 March
GDPI	Gross Direct Premium Income
GenAI	Generative artificial intelligence
IRDAI	Insurance Regulatory and Development Authority of India
MTPL	Motor third-party liability
NEP	Net earned premium
PAT / PBT	Profit after tax / profit before tax
RoAE	Return on average equity
R_f, R_m	Risk-free rate (6% a year) and market return
σ, Σ	Standard deviation and covariance matrix
VaR	Value at Risk (historical, 5%, monthly)
₹ crore	Indian rupees, ten million

1. Introduction

1.1 Background and Motivation

An insurer collects premiums today and pays uncertain claims later. In the meantime, it invests the money it holds and keeps enough capital to survive bad years. Each of these activities depends on analysis. Underwriters need to judge how likely a policy is to produce a claim, claims teams need to decide which cases deserve a closer look, investment teams need to balance return against volatility, and regulators need to see that capital is adequate. Financial and risk analytics supplies the methods for these tasks: ratio analysis, risk measurement, portfolio theory, statistical classification, anomaly detection and decision-support tools.

The Indian general insurance market makes these questions practical. ICICI Lombard's 2025-26 annual report states that industry premium grew by 9.3% in FY2026, while the industry combined ratio rose to 119.3% in the first nine months of the year (ICICI Lombard General Insurance Company Limited [ICICI Lombard], 2026). Growth is coming with weak underwriting margins, so the quality of risk selection and cost control matters more than premium volume.

1.2 Scope and Structure of the Report

The report is an individual sector analysis under the Insurance and Risk Analytics option of the assignment brief. It brings together four analyses that use different data and answer different questions (Table 2). They are kept apart throughout: stock-market results say nothing about the accuracy of the claim model, and French policy data say nothing about ICICI Lombard's customers. Before any analysis, the CIA-4 workbook was checked against the audited statements and its errors were corrected (Section 5.2), and the claim-risk code was extended and re-run (Chapter 8). All figures in this report come from the corrected workbook and from the outputs of the enhanced code.

Table 2: Analytical components, evidence and chapters

Component	Evidence used	Unit of analysis	Ch.
Financial statement analysis	Corrected CIA-4 workbook; ICICI Lombard annual reports FY2024 to FY2026	ICICI Lombard, 3 years	6
Market risk and portfolios	Workbook monthly price data and Portfolio Analysis sheet; Python visualisation code	104 monthly returns	7
Claim-risk prediction	Enhanced Python pipeline on freMTPL2freq	678,013 French policies	8
Anomaly screening	Two Isolation Forest screens in the same pipeline	678,013 policies	9
Prediction dashboard	ipywidgets predictor in the enhanced code	Single policy	10

Source: Author.

2. Insurance Sector Overview

2.1 Structure and Business Model

Insurance pools many independent exposures so that the total loss becomes predictable even though individual losses are not (Rejda et al., 2020). The three listed companies studied in Chapter 7 cover the three parts of the sector. ICICI Lombard is a general insurer writing mostly one-year motor, health, fire, marine and liability cover. SBI Life is a life insurer with long-term savings and protection contracts. GIC Re is a reinsurer that takes on large and catastrophe risks from other insurers. All three are regulated by IRDAI.

A general insurer has two sources of profit. Underwriting profit is earned premium minus claims, commission and operating expenses. Investment income comes from investing premiums until claims are paid, together with shareholders' funds. The combined ratio measures the first source: above 100% means the underwriting account loses money and profit depends on investment income.

2.2 Revenue, Costs and Capital

Premium income is reported as GDPI (business written), net written premium (after reinsurance) and net earned premium. The main costs are claims incurred, commission and expenses of management. Under the IRDAI expenses of management regulations of 2023, commission is part of total expenses of management, so the commission ratio and the expense ratio overlap and should not be added together (ICICI Lombard, 2024). Capital adequacy is measured by the solvency ratio, which must stay above 1.50 times. From 1 October 2024 long-term non-motor policies are accounted for on a 1/n basis, spreading premium over the policy term. This affects the comparison of FY2025 and FY2026 with earlier years (ICICI Lombard, 2026).

2.3 Major Risks

Table 3: Main insurer risks and where they are addressed

Risk	Description	Addressed in
Underwriting risk	Premiums prove too low for claims and expenses	Ch. 6 (combined ratio), Ch. 8
Reserving risk	Claim provisions prove insufficient	Ch. 6 (provisions), discussed only
Catastrophe risk	Clustered losses from natural events	Ch. 7 (reinsurer volatility)
Market risk	Falls in investment values	Ch. 6 (FVCA), Ch. 7
Liquidity risk	Difficulty paying claims when due	Ch. 6 (cash flow)
Fraud and data risk	Inflated claims and wrong records	Ch. 9 (screening only)
Model risk	Decisions based on a wrong or misused model	Ch. 8 to 10, 13

Risk	Description	Addressed in
Regulatory risk	Changes in expense, accounting or capital rules	Ch. 2, 6

Source: Author's classification informed by Rejda et al. (2020) and ICICI Lombard (2026).

2.4 Indian Context and the Role of Analytics

ICICI Lombard held an 8.5% share of industry GDPI and 12.9% of the private sector in FY2026, issued 39.3 million policies and processed 3.4 million claims (ICICI Lombard, 2026). The same report mentions pricing pressure in fire insurance, a large gap between economic and insured catastrophe losses, and the Insurance Laws (Amendment) Act, 2025, which allows 100% foreign direct investment. In this setting analytics supports pricing and underwriting through risk scoring, claims management through triage and review queues, solvency and ratio monitoring, and investment decisions through risk-adjusted measures. The company itself describes underwriting supported by advanced analytics and improved risk selection models (ICICI Lombard, 2026).

3. Research Objectives

3.1 Objectives

The overall objective is to show how financial and risk analytics can support decisions in the insurance sector, using real data and stating clearly what each analysis can and cannot support. The specific objectives are:

1. to evaluate ICICI Lombard's growth, profitability, underwriting performance, solvency and cash flow for FY2024 to FY2026;
2. to measure the return, volatility, beta and correlation of three listed Indian insurers against the NIFTY 50;
3. to compare equal-weight, maximum-Sharpe and minimum-variance portfolios of these insurers;
4. to build and evaluate a claim-occurrence model and an exposure-based claim-frequency model;
5. to screen policy records for unusual observations and explain how the flags should be used;
6. to show how an interactive predictor can present model outputs to decision-makers; and
7. to derive recommendations that follow from the evidence.

4. Literature Review and Theoretical Framework

4.1 Financial Statement and Insurance Ratio Analysis

Horizontal analysis compares each line with the previous year and vertical analysis expresses each line as a share of a base (Penman, 2013). Insurers need their own ratios because their balance sheets consist mostly of investments and technical liabilities. IRDAI's analytical ratios, published in every annual report, include the claims ratio, commission ratio, expense ratio, combined ratio, operating profit ratio and solvency ratio. Piotroski (2000) proposed a nine-signal score to separate improving from weakening value firms. Some of its signals, such as gross margin and current ratio, do not exist for insurers, so the score has to be adapted.

4.2 Risk, Return and Portfolio Theory

Markowitz (1952) showed that portfolio risk depends on the covariances between assets, so combining assets that are not perfectly correlated lowers risk. Under the CAPM only systematic risk, measured by beta, earns a premium (Sharpe, 1964; Lintner, 1965), although beta alone explains returns poorly in practice (Fama & French, 2004). The Sharpe ratio scales excess return by total volatility (Sharpe, 1966) and the Treynor ratio scales it by beta (Treynor, 1965). Optimised weights are sensitive to errors in estimated means (Michaud, 1989), and a simple equal-weight portfolio is often hard to beat out of sample (DeMiguel et al., 2009).

4.3 Claim-Risk Modelling and Classification

Non-life pricing models claim frequency with Poisson regression and treats exposure as an offset (Ohlsson & Johansson, 2010; Denuit et al., 2019). The freMTPL2freq dataset is the standard public benchmark for such models (Noll et al., 2018; Wüthrich & Merz, 2023). Its Bonus-Malus variable reflects past claims (Lemaire, 1995). Predicting whether a policy has a claim is an imbalanced classification problem, where accuracy is misleading and precision, recall and Average Precision are more informative than ROC-AUC alone (Saito & Rehmsmeier, 2015; Davis & Goadrich, 2006). Logistic regression is the interpretable baseline (Hosmer et al., 2013). Random forests average many decision trees (Breiman, 2001), and gradient boosting adds trees one at a time to correct the errors of the previous ones (Friedman, 2001). Calibration can be checked with the Brier score (Brier, 1950) and corrected with isotonic regression (Zadrozny & Elkan, 2002). Bootstrap resampling gives confidence intervals for test metrics (Efron & Tibshirani, 1993).

4.4 Anomaly Detection and Decision Support

Anomaly detection looks for records that do not fit the usual pattern (Chandola et al., 2009). Isolation Forest separates records by random splits; unusual records are isolated in fewer splits (Liu et al., 2008). Insurers use such scores to prioritise claims for review (Viaene et al., 2002; Phua et al., 2010), but without fraud labels a score cannot be validated as a fraud detector. Decision-support systems combine data, models and an interface to assist human judgement (Power, 2002), and predictive models add value when tied to a real decision and its costs (Shmueli & Koppius, 2011).

4.5 AI, Generative AI and Responsible Analytics

Eling et al. (2022) map AI uses along the insurance value chain from marketing to claims. The OECD (2020) notes efficiency gains together with risks of opacity and unfair treatment. Generative AI is built on large foundation models that can produce fluent but wrong output (Bommasani et al., 2021). Model-risk guidance asks for independent validation, documentation and monitoring (Board of Governors of the Federal Reserve System, 2011), and insurance-specific principles stress transparency, fairness and human oversight (European Insurance and Occupational Pensions Authority [EIOPA], 2021; National Institute of Standards and Technology [NIST], 2023). Data leakage gives optimistic results (Kaufman et al., 2012). Permutation importance and partial dependence describe how a model uses its inputs (Breiman, 2001; Friedman, 2001; Fisher et al., 2019).

4.6 Theoretical Framework

Table 5: Theory, analysis and evidence

Theory or concept	Analysis	Evidence produced
Ratio analysis; IRDAI ratios; Piotroski (2000)	Ch. 6	Growth, margins, combined ratio, solvency, F-score
Portfolio theory; CAPM; Sharpe and Treynor	Ch. 7	Risk, beta, portfolio performance, opportunity set
Imbalanced classification; gradient boosting; Poisson frequency	Ch. 8	Test metrics, calibration, bands, deviance
Isolation Forest	Ch. 9	Flags, reasons, stability
Decision support; model governance	Ch. 10, 13	Predictor functions; recommendations

5. Data Sources and Preparation

5.1 Dataset Inventory

Table 6: Datasets used

Dataset	Source	Period or size	Used in
ICICI Lombard P&L, Balance Sheet, Cash Flow, Ratio and Piotroski sheets	Corrected CIA-4 workbook, based on the integrated annual reports	FY2024 to FY2026, ₹ crore	Ch. 6
Monthly Price and Return Data; stock sheets; Portfolio Analysis	Corrected CIA-4 workbook (price source not recorded)	Oct 2017 to Jun 2026; 104 returns	Ch. 7
freMTPL2freq 78.csv	French MTPL dataset (Dutang & Charpentier, n.d.), renamed columns	678,013 policies, 12 columns	Ch. 8 to 10

.

5.2 Financial Statement Data and Workbook Validation

Every input in the financial sheets was compared with the audited statements. All 23 Profit and Loss lines and all Balance Sheet lines for the three years agree with the annual reports, as do GDPI, PAT, EPS, RoAE, the combined ratio and the solvency ratio (ICICI Lombard, 2024, 2025, 2026). Calculations, however, contained several errors, which were corrected before this report was written. Table 7 summarises the main corrections; the workbook's Validation Log sheet lists all 36 checks.

Table 7: Main corrections made to the CIA-4 workbook

Sheet	Issue found	Correction
Ratio Analysis	FY2024 operating profit ratio of 16% was the FY2023 value	Set to 11.2% as published
Ratio Analysis	IRDAI ratios rounded to whole percentages; solvency stored as text	Exact published values; solvency as numbers
Stock sheets	CAPM formula was $R_f \times \beta + (R_m - R_f)$	Changed to $R_f + \beta(R_m - R_f)$
ICICI Lombard sheet	Used 105 returns while other sheets used 104	Aligned to Nov 2017 to Jun 2026
Cash Flow Summary	Whole-crore values; FY2024 opening cash wrong (204)	Exact audited values; opening cash 203.13
Piotroski sheet	Wrong references; FY2024 averages missing	Formulas fixed; FY2023 assets added
P&L and Balance Sheet	Growth on negative bases misleading; one wrong base cell	Sign-safe growth formula; base corrected
Portfolio Analysis	Statistics typed as rounded numbers	Replaced by formulas; same results

5.3 Monthly Stock Data

The monthly sheet holds closing prices for ICICI Lombard, GIC Re, SBI Life and the NIFTY 50 from October 2017 to June 2026. Simple returns, $P_t / P_{t-1} - 1$, give 104 aligned monthly returns. The NIFTY values match the published month-end closes at every date checked, but the June 2026 value (23,382.60) is below the 30 June 2026 close of about 23,865, so the last

row appears to be a mid-month figure. The prices also appear to exclude dividends. The risk-free rate is 6% a year. The Portfolio Analysis sheet uses the historical NIFTY return as the market return in the CAPM, while the stock sheets use an assumed 15%.

5.4 The freMTPL2freq Claim Data

The claim model uses 678,013 French motor third-party liability policies, each with an exposure period and a claim count. The dataset is the standard public benchmark for claim-frequency models (Noll et al., 2018). It is not Indian data, and its Bonus-Malus scale, regions and brand codes do not exist in Indian portfolios. The target is Claim_Flag = 1 when the policy has at least one claim; 34,060 policies (5.02%) do. The nine predictors are vehicle power, vehicle age, driver age, Bonus-Malus score, population density, vehicle brand (11 codes), fuel type, area (6) and region (21). Policy ID, claim count and exposure are not used as predictors in the classification model. The frequency model uses claims and exposure, as explained in Section 8.6.

The validation step found no impossible values, so all 678,013 rows were kept. One density value and one area value were blank and are handled inside the model pipeline. Review flags were raised for 97 vehicles older than 50 years, 1,224 exposures above one year and 7 policies with more than five claims; these rows were kept. The file is ordered with claimants concentrated at the top (31.7% of the first 5% of rows have a claim), so rank-based deciles that break ties by row order give distorted charts. The enhanced code therefore uses value-based bands.

6. Financial Statement and Insurance Ratio Analysis

This chapter uses the corrected workbook. Horizontal analysis uses the sign-safe formula (current - previous) / |previous|, with 'n.m.' (not meaningful) where the base is zero or the sign changes. Vertical analysis of the Profit and Loss Account uses Total Income (A) as the base, which is the sum of segment operating profit and the shareholders' investment and other income.

6.1 Growth and Profitability

Table 8: Headline indicators, FY2024 to FY2026

Indicator	FY2024	FY2025	FY2026	FY25 chg	FY26 chg
GDPI (₹ crore)	24,776	26,833	28,712	+8.30%	+7.00%
Total Income (A) (₹ crore)	2,755.92	3,331.13	3,716.97	+20.87%	+11.58%
Profit before tax (₹ crore)	2,555.18	3,321.31	3,658.93	+29.98%	+10.17%
Profit after tax (₹ crore)	1,918.59	2,508.29	2,771.94	+30.74%	+10.51%
Basic EPS (₹)	39.03	50.74	55.74	+30.00%	+9.85%
Return on average equity	17.2%	19.1%	17.8%	+1.9 pp	-1.3 pp
PAT / GDPI	7.74%	9.35%	9.65%	+1.61 pp	+0.30 pp

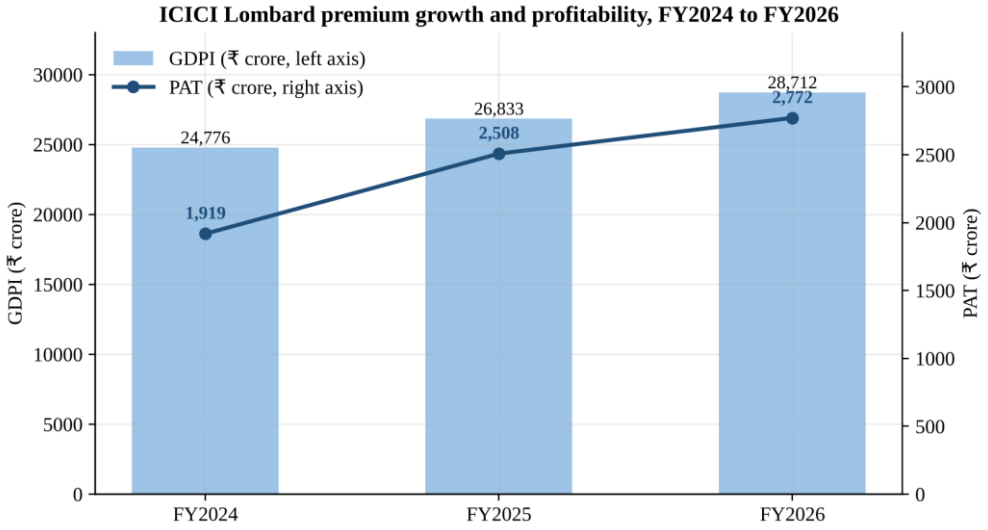


Figure 1: GDPI and PAT of ICICI Lombard (₹ crore)

Interpretation: Premium growth slowed from 8.3% to 7.0%, while profit grew faster than premium in both years, so PAT per rupee of GDPI rose from 7.74 to 9.65 paise (Figure 1). The annual report links the slower FY2026 growth to a choice of profitable business in a competitive market (ICICI Lombard, 2026). RoAE fell from 19.1% to 17.8% in FY2026 because reserves grew by 18.4%, faster than profit (10.5%).

6.2 Horizontal and Vertical Analysis

Table 9: Horizontal analysis of selected P&L lines (₹ crore)

Particulars	FY2024	FY2025	FY2026	FY25 chg	FY26 chg
Operating profit: Fire	503.67	783.31	630.51	+55.52%	-19.51%
Operating profit: Marine	27.21	-7.20	-73.62	n.m.	-922.50%
Operating profit: Miscellaneous	1,375.02	1,490.81	1,966.84	+8.42%	+31.93%
Total segment operating profit	**1,905.90**	**2,266.92**	**2,523.73**	**+18.94%**	**+11.33%**
Interest, dividend and rent	677.18	804.83	887.41	+18.85%	+10.26%
Profit on sale of investments	170.92	225.17	275.87	+31.74%	+22.52%
Total (B): provisions and other expenses	200.74	9.82	58.04	-95.11%	+491.04%
Profit after tax	**1,918.59**	**2,508.29**	**2,771.94**	**+30.74%**	**+10.51%**

Source: Corrected CIA-4 workbook, 'Profit & Loss Account'.

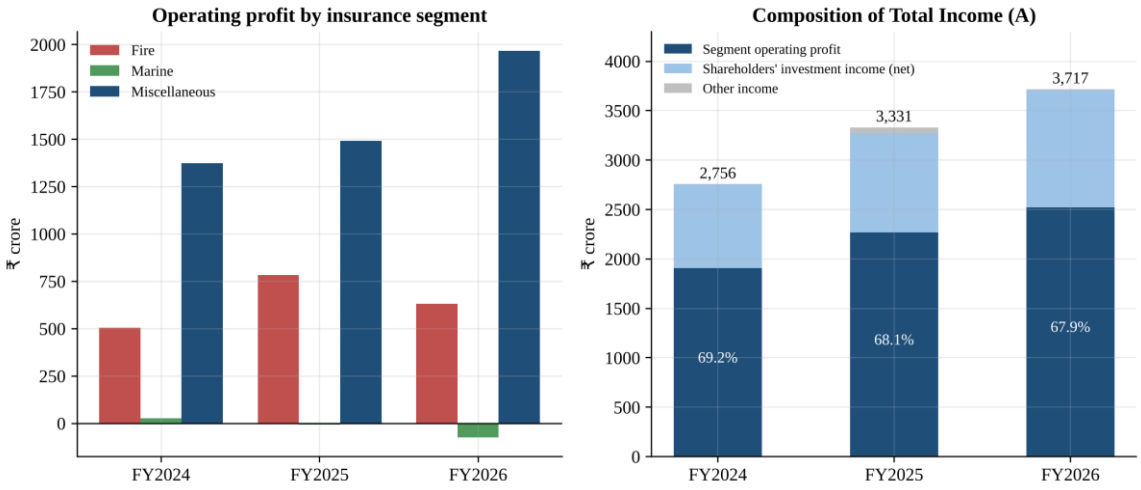


Figure 2: Segment operating profit and composition of Total Income (A)

Interpretation: The Miscellaneous segment (motor, health and liability) drove FY2026, with operating profit up 31.9%, while Fire fell by 19.5% and Marine moved from a small profit to a loss of ₹73.62 crore. Part of FY2025's 30.7% profit growth did not come from operations. Total (B) fell from ₹200.74 crore to ₹9.82 crore, mainly because a ₹95.82 crore provision for diminution in investment value in FY2024 became a write-back of ₹40.54 crore. This ₹190.92 crore swing explains 24.9% of the rise in PBT, so FY2026's 10.5% growth is the better guide to underlying momentum. In the vertical analysis, segment operating profit is 68 to 69% of Total Income in each year and interest, dividend and rent about 24% (Figure 2). Since segment profit itself includes investment income on policyholders' funds, the dependence on investment returns is higher than these shares suggest.

6.3 Underwriting Ratios and Solvency

Table 10: Insurance operating ratios and solvency

Ratio	FY2024	FY2025	FY2026
Combined ratio (1/n basis)	103.3%	102.8%	103.4%
Combined ratio ('n' basis)	n/a	102.6%	102.4%
Net incurred claims / NEP	70.8%	70.6%	71.1%
Expenses of management / GDPI	29.8%	30.8%	32.0%
Net commission ratio	17.0%	18.5%	19.0%
Operating profit ratio	11.2%	11.4%	11.0%
Solvency ratio (minimum 1.50)	2.62×	2.69×	2.67×

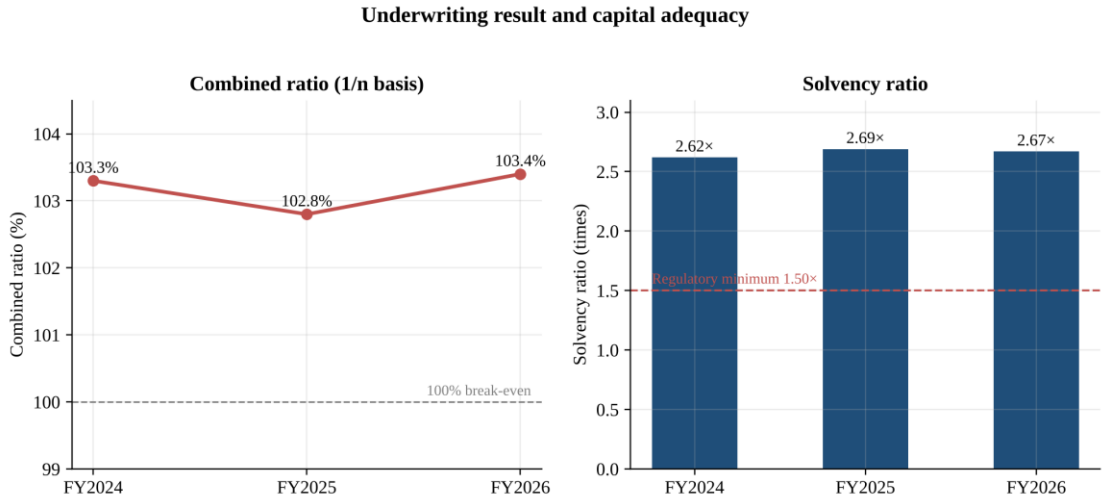


Figure 3: Combined ratio and solvency ratio

.

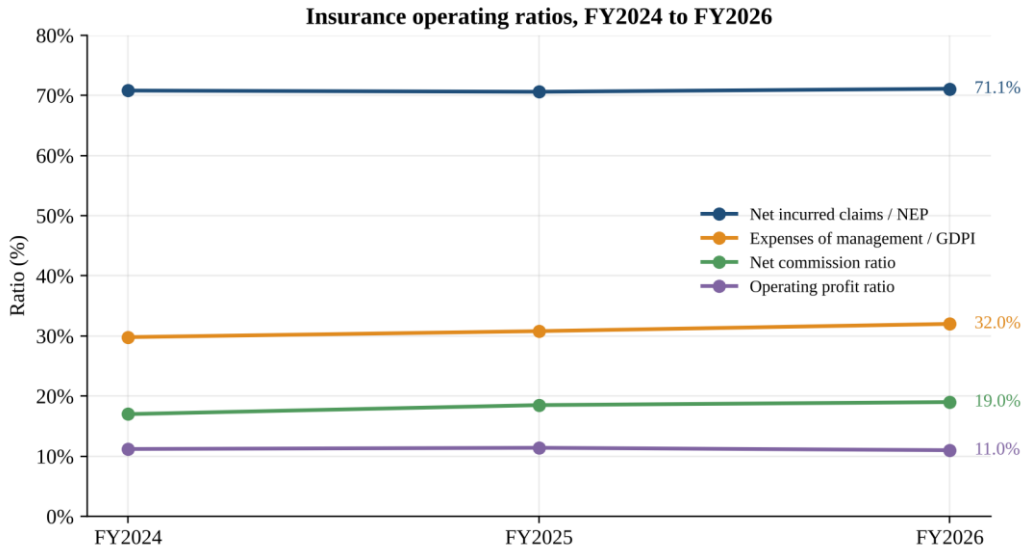


Figure 4: Insurance operating ratios, FY2024 to FY2026

.

Interpretation: For every ₹100 of premium, ICICI Lombard spent about ₹103 on claims, commission and expenses, and earned its profit from investing the float (Figure 3). This is still far better than the industry's 119.3% for the first nine months of FY2026. On the 'n' basis, which removes the 1/n accounting change, the ratio improved slightly from 102.6% to 102.4%, so the reported rise to 103.4% is partly an accounting effect. The claims ratio was stable at about 71%, while the commission ratio rose from 17.0% to 19.0% and the expense ratio from 29.8% to 32.0% (Figure 4). Cost drift, not claims, is pushing the combined ratio up. Solvency of about 2.7 times is roughly 1.8 times the regulatory floor.

6.4 Balance Sheet and Cash Flow

Table 11: Balance-sheet and cash-flow summary (₹ crore)

Item	FY2024	FY2025	FY2026
Share capital and reserves	11,959.78	14,303.10	16,845.91
Fair value change accounts (total)	989.55	680.74	-774.47
Borrowings	35.00	0.00	0.00
Investments (shareholders + policyholders)	48,907.24	53,507.77	58,421.27
Net current assets	-36,915.68	-39,494.73	-43,305.25
Total assets	63,308.29	69,020.26	76,110.94
Net cash from operating activities	2,406.77	1,147.30	2,621.97
Net cash from investing activities	-1,920.74	-1,137.26	-1,732.72
Net cash from financing activities	-354.59	-257.05	-335.54
Cash and cash equivalents at year end	334.58	87.57	640.33
CFO / PAT	1.25×	0.46×	0.95×

Source: Corrected CIA-4 workbook ('Balance Sheet', 'Cash Flow Summary', 'PIOTROSKI-SCORE'); ratios computed by the author.

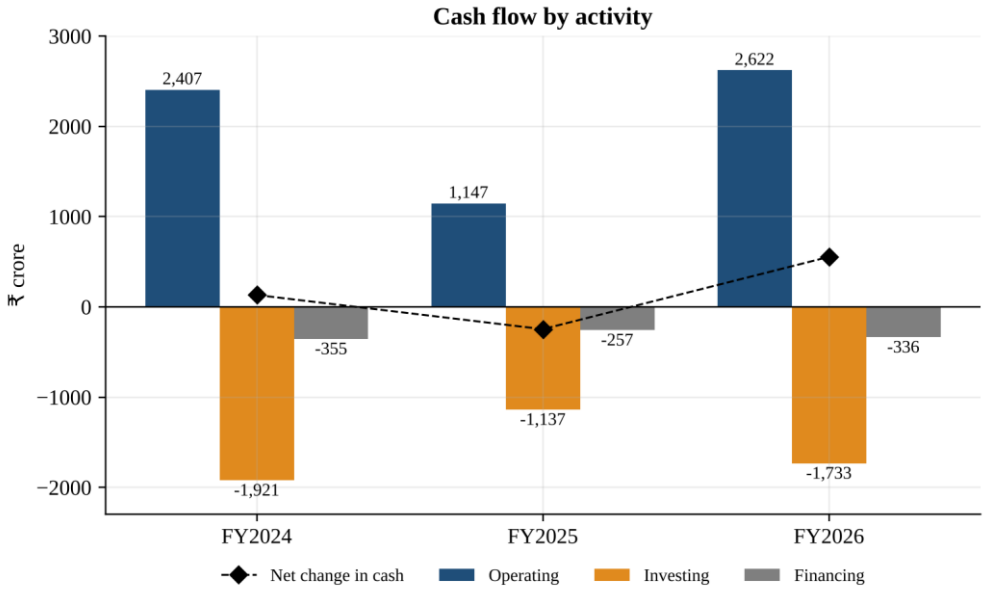


Figure 5: Cash flow by activity (₹ crore)

Interpretation: Investments grew by about 9% a year to ₹58,421 crore, three-quarters of it policyholders' funds. Net current assets are deeply negative because current liabilities and provisions are mostly claim and unearned-premium reserves, while the assets that back them are held as investments. For an insurer this is a normal structure, not a sign of distress. Two points matter more. Borrowings were repaid in FY2025. Both fair value change accounts turned negative in FY2026 (a total of -₹774.47 crore against +₹680.74 crore a year earlier), which shows unrealised market losses on the investment book. Operating cash flow was positive every year but dipped to 0.46 times PAT in FY2025 (Table 11, Figure 5). Year-end cash in FY2026 (₹640.33 crore) is lower than 'cash and bank balances' in the balance sheet (₹986.78 crore) because some deposits are not treated as cash equivalents.

6.5 Piotroski-Style Score

Table 12: Piotroski criteria after correction

Criterion	FY2025	FY2026	Basis
1. Positive ROA	1	1	ROA 3.79% and 3.82%
2. Positive operating cash flow	1	1	CFO positive
3. Improving ROA	1	1	3.24% to 3.79% to 3.82%
4. CFO above PAT	0	0	CFO below PAT
5. No increase in leverage	1	1	Debt repaid; nil in FY2026
6. Improving current ratio	n/a	n/a	Not applicable to insurers
7. No new equity	0	0	Small ESOP allotments
8. Improving gross margin	n/a	n/a	Not applicable to insurers
9. Improving asset turnover	0	0	0.419 to 0.406 to 0.396
Score (applicable criteria)	4 of 7	4 of 7	57%: 'Weak, investigate'

Interpretation: ICICI Lombard meets 4 of the 7 applicable criteria in both years. It passes on profitability and cash flow but fails on accruals (cash flow below profit), small equity issues and falling asset turnover, since total assets grew faster than premium. The 'weak' label should be read with care. The score was designed for industrial value stocks, and two of its criteria cannot be measured for an insurer. The direct evidence in Sections 6.1 to 6.4, which shows rising profit, stable solvency and no debt, is more informative.

6.6 Implications

ICICI Lombard grows more slowly than the market, keeps its combined ratio near 103% and earns a 17 to 19% RoAE mainly from investment income on a growing float. Its main financial exposures are therefore investment-market risk, already visible in the FY2026 fair value change

accounts; the steady rise in commission and expense ratios; and the concentration of profit in the Miscellaneous segment while Marine makes a loss. Better risk selection, the subject of Chapter 8, is one of the few levers that can bring the combined ratio below 100%.

7. Stock Market Risk and Portfolio Analysis

7.1 Method

The analysis uses 104 monthly returns from November 2017 to June 2026. The following formulas are implemented in the workbook and in the Python visualisation code; Σ is the monthly covariance matrix and the risk-free rate is 6% a year.

$$R = 12 \times mean(r_t); \quad \sigma = \sqrt{12} \times s(r_t); \quad \beta = Cov(r_i, r_m) / Var(r_m) \tag{7.1}$$

$$R_p = \Sigma\, w_i R_i; \quad \sigma_p = \sqrt{(12 \times w^T \Sigma w)} \tag{7.2}$$

$$Sharpe = (R_p - R_f) / \sigma_p; \quad Treynor = (R_p - R_f) / \beta_p; \quad CAPM = R_f + \beta(R_m - R_f) \tag{7.3}$$

Skewness, kurtosis, historical 5% VaR, maximum drawdown and compound annual growth were added by the author from the same data to describe downside risk.

7.2 Return and Risk of the Individual Stocks

Table 13: Return and risk of the insurance stocks and NIFTY 50

Metric	ICICI Lombard	GIC Re	SBI Life	NIFTY 50
Annualised return	13.57%	7.47%	14.83%	10.91%
CAGR	11.57%	-1.95%	11.98%	9.88%
Annualised volatility	22.96%	44.35%	25.93%	16.89%
Beta	0.834	1.205	0.939	1.000
Correlation with NIFTY 50	0.613	0.459	0.611	1.000
Sharpe ratio	0.330	0.033	0.340	0.291
Treynor ratio	0.091	0.012	0.094	0.049
Skewness / excess kurtosis	0.15 / -0.04	0.76 / 2.69	-0.73 / 2.57	-0.95 / 5.25
Historical 5% monthly VaR	-9.89%	-17.37%	-11.23%	-5.86%
Maximum drawdown	-32.9%	-73.8%	-35.5%	-29.3%

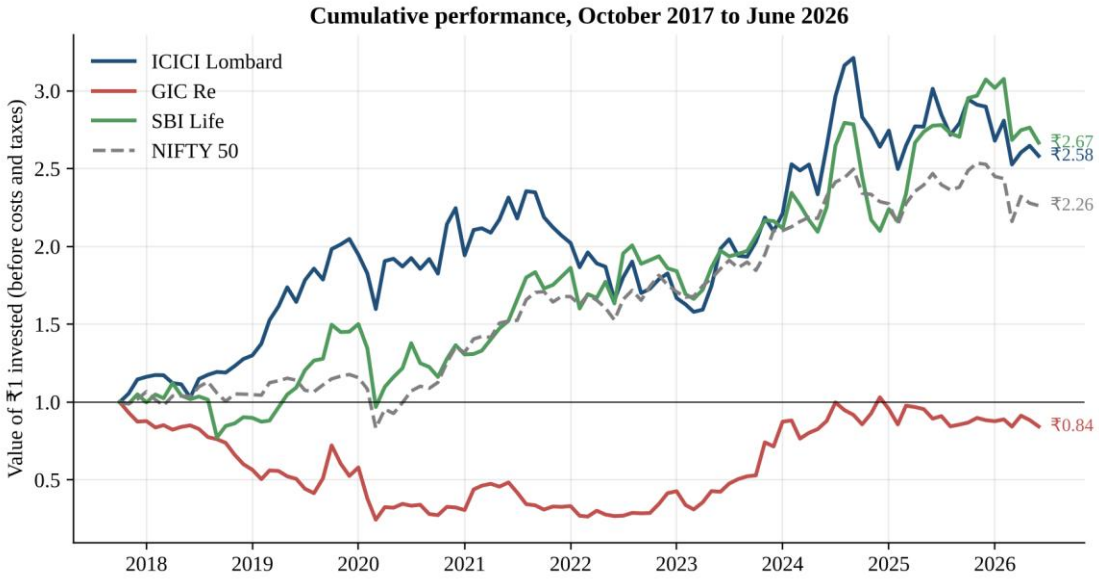


Figure 6: Growth of ₹1 invested in October 2017

Source: Corrected workbook monthly data; supplied visualisation code chart 01.

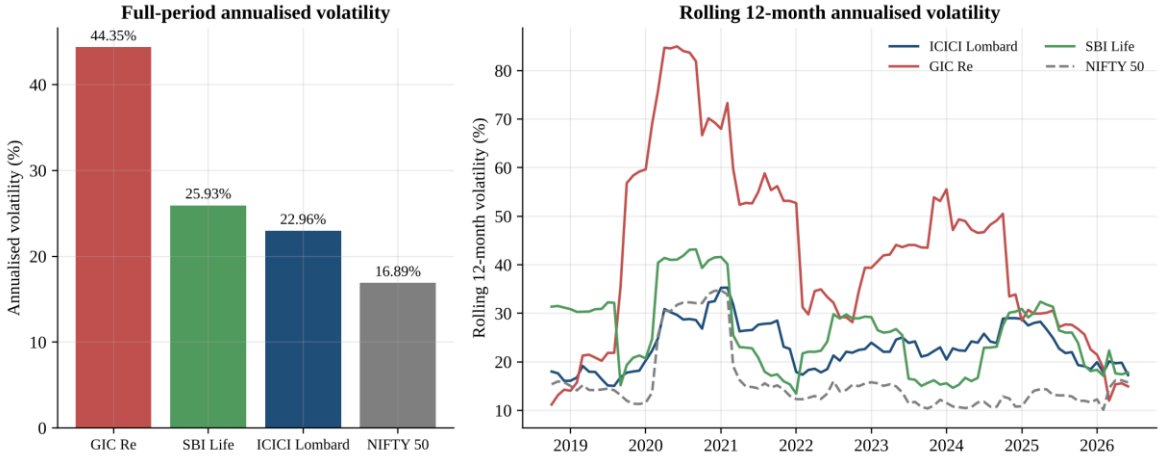


Figure 7: Full-period and rolling 12-month annualised volatility

Interpretation: ₹1 grew to ₹2.67 in SBI Life and ₹2.58 in ICICI Lombard, ahead of ₹2.26 in the NIFTY 50, but fell to ₹0.84 in GIC Re (Figure 6). GIC Re lost 73.8% from its peak to March 2020 and, despite a strong recovery to ₹444.25 in December 2024, ended below its starting price. Its volatility of 44.35% is almost twice SBI Life's, consistent with a reinsurer's exposure to large and catastrophe losses (Figure 7). Volatility also changes over time: GIC Re's 12-month volatility peaked at 85% in June 2020 and fell to 15% by June 2026. A high arithmetic return can hide poor compounding. GIC Re's annual average return of 7.47% sits alongside a negative CAGR of -1.95%, because high volatility drags down compound growth by roughly $\sigma^2/2$. Returns are also not normal: SBI Life and the index are negatively skewed with fat tails, so volatility understates downside risk (Table 13).

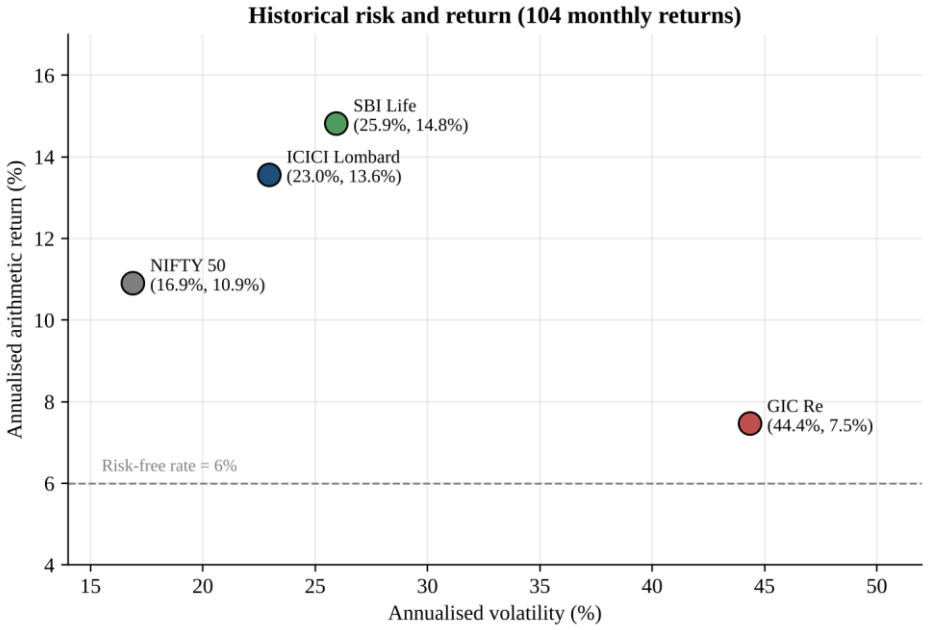


Figure 8: Historical risk and return

Interpretation: SBI Life and ICICI Lombard lie above the index in risk-adjusted terms, with Sharpe ratios of 0.340 and 0.330 against 0.291. GIC Re lies far to the right with a return only 1.5 percentage points above the risk-free rate, so it is dominated by the other two (Figure 8).

7.3 Correlation, Beta and CAPM

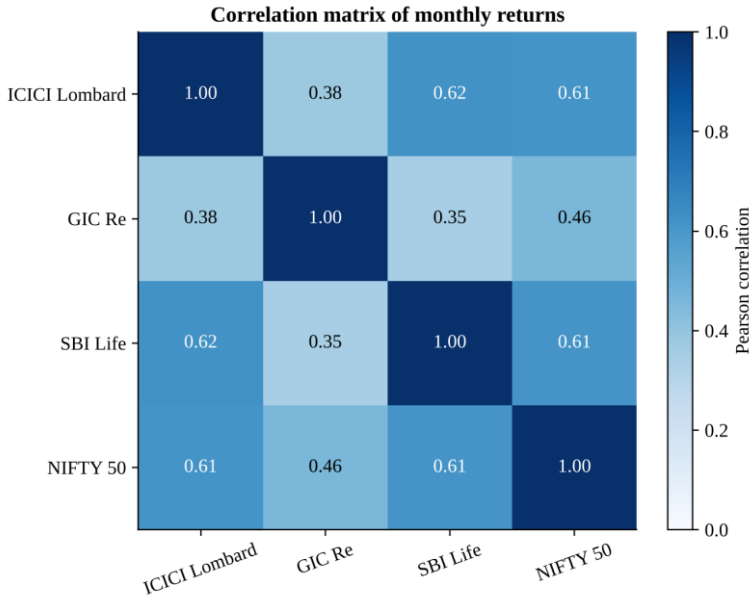


Figure 9: Correlation matrix of monthly returns

Table 14: Beta and CAPM required return

Item	ICICI Lombard	GIC Re	SBI Life
Beta	0.834	1.205	0.939
CAPM, R_m = historical NIFTY 10.91%	10.10%	11.92%	10.61%
Actual return minus CAPM (historical R_m)	+3.47 pp	-4.45 pp	+4.22 pp
CAPM, R_m = assumed 15%	13.51%	16.84%	14.45%

Interpretation: Correlations between the insurers are moderate, from 0.35 (GIC Re and SBI Life) to 0.62 (ICICI Lombard and SBI Life), so there is room to diversify (Figure 9). ICICI Lombard is the most defensive stock with a beta of 0.834, which fits a business built on compulsory and renewable cover. GIC Re is the only stock with a beta above 1. Against the CAPM with the historical market return, ICICI Lombard and SBI Life earned 3.5 to 4.2 points more than their systematic risk required, while GIC Re earned 4.45 points less (Table 14). These are in-sample differences, not evidence of lasting outperformance.

7.4 Portfolio Construction and Performance

Table 15: Portfolio weights and performance (annualised, R_f = 6%)

Portfolio	ICICI L.	GIC Re	SBI Life	Return	Risk	Sharpe	Beta
A: Equal weight	33.33%	33.33%	33.33%	11.96%	24.66%	0.242	0.992
B: Maximum Sharpe	49.64%	0.00%	50.36%	14.20%	22.04%	0.372	0.887
C: Minimum variance	62.63%	5.08%	32.29%	13.67%	21.66%	0.354	0.887
NIFTY 50 (reference)				10.91%	16.89%	0.291	1.000

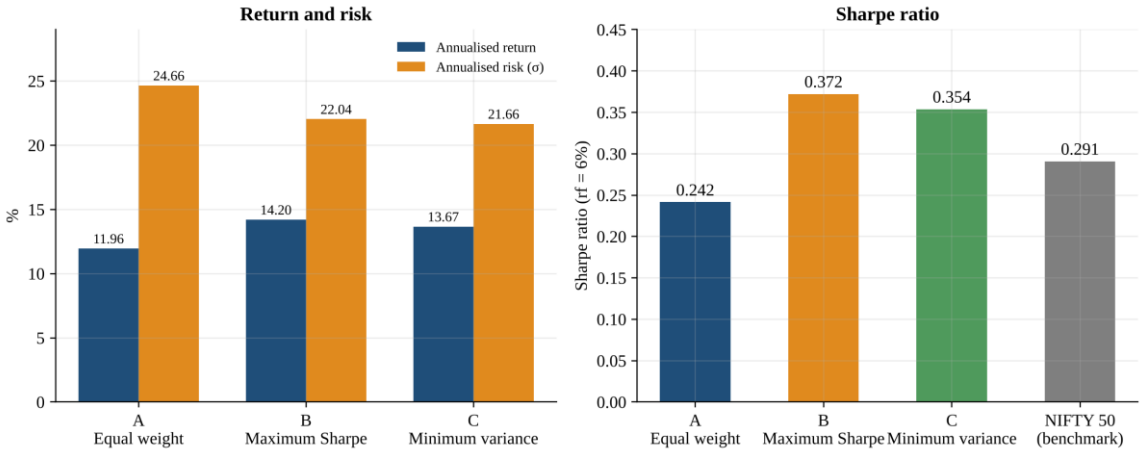


Figure 10: Portfolio return, risk and Sharpe ratio

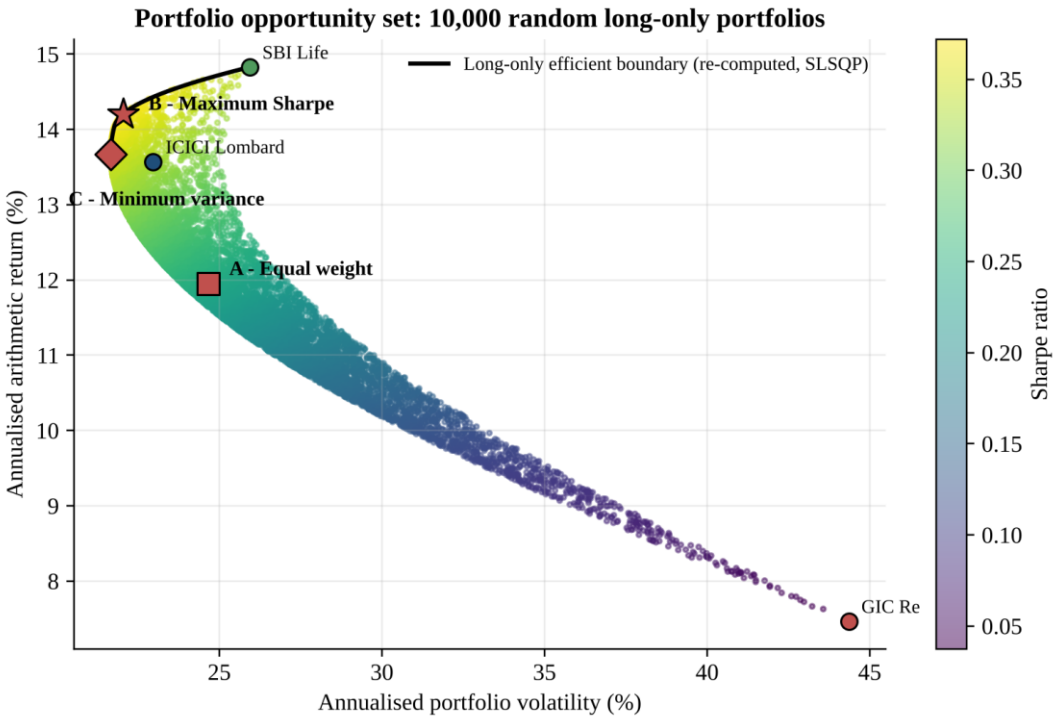


Figure 11: Opportunity set of 10,000 random portfolios with the efficient boundary

Interpretation: Diversification lowers risk even within one sector: Portfolio A's risk (24.66%) is well below the average volatility of its three stocks (31.08%). Because A gives a third of the money to GIC Re, its Sharpe ratio (0.242) is still below the index. Portfolio B drops GIC Re, splits capital almost equally between ICICI Lombard and SBI Life, and earns 2.24 points more than A with 2.62 points less risk (Table 15, Figure 10). Portfolio C keeps 5% in GIC Re because its low correlation reduces variance, and reaches the lowest risk (21.66%). The supplied code calls its random-portfolio scatter an efficient frontier. Strictly, it is the set of feasible portfolios. The solid line in Figure 11 is the true long-only boundary, and it confirms that B is the tangency portfolio and C the minimum-variance point.

7.5 Implications

Over this period the general insurer and the life insurer rewarded investors better than the reinsurer, and excluding GIC Re did more for risk-adjusted return than equal weighting. The result depends on the sample window. Over the last 36 months alone GIC Re returned 28.3% a year against 11.2% for ICICI Lombard and 12.5% for SBI Life, so an optimisation on that window would look very different (Michaud, 1989). Portfolio B also holds only two stocks from one sector, so the optimised weights are best treated as a diagnostic of past performance rather than an allocation to follow.

8. Insurance Claim-Risk Prediction Framework

8.1 Problem Definition and Target

The framework has two tracks. The **classification track** predicts claim occurrence: the probability that a policy records at least one claim during its observed period (Claim_Flag). Exposure is not an input, so this probability refers to the typical observation period in the data (median 0.49 years). The **frequency track** predicts claims per policy-year, using exposure as the weight in a Poisson model, which is the standard actuarial approach (Ohlsson & Johansson, 2010). Neither track predicts claim cost, fraud or premium.

8.2 Data Inspection

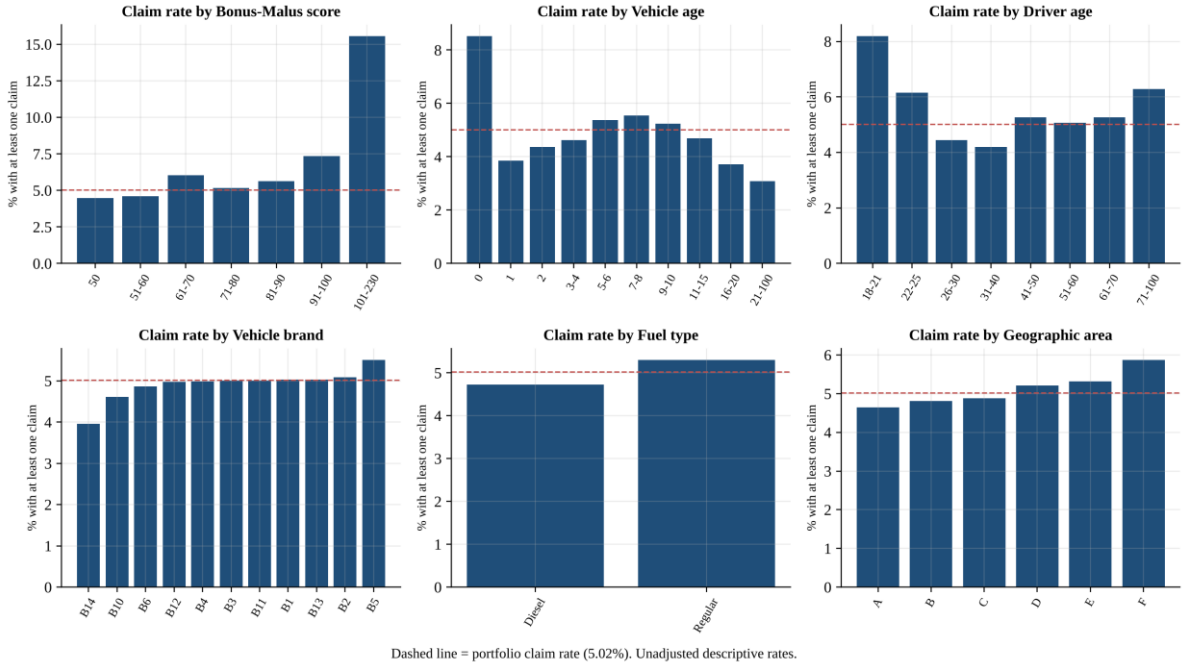


Figure 12: Observed claim rate by predictor, all 678,013 policies

Interpretation: Claim rates rise with Bonus-Malus (4.46% at the minimum score of 50 against 15.46% for scores of 101 to 150) and are highest for drivers aged 18 to 21 (8.19%). Brand-new vehicles show 8.52% even though their median exposure is only 0.18 years (Figure 12). Chi-square tests on value-based bins find every predictor significantly associated with claims after a Bonferroni correction, but all effects are small: Cramér's V ranges from 0.007 (vehicle brand) to 0.047 (Bonus-Malus). No single variable separates claimants from non-claimants, so modest discrimination should be expected.

8.3 Preprocessing, Partitioning and Models

The data are split by stratified sampling into training (406,807 policies), validation (135,603) and test (135,603) sets, each with a claim rate of 5.02%. There is no policy date, so an out-of-time test is not possible. All preprocessing is fitted on training rows only. For logistic regression and random forest, numeric inputs are median-imputed, density is log-transformed and categories are one-hot encoded. For gradient boosting, categories are passed directly to the model's native categorical handling and missing values are handled internally. The model, the calibration decision and the alert threshold are all chosen on the validation set, and the test set is scored once (Pedregosa et al., 2011).

$$\text{Logistic regression: } P(y = 1 \mid x) = 1 / (1 + \exp(-(\beta_0 + \beta^T x))) \quad (8.1)$$

$$\text{Gradient boosting: } F_m(x) = F_{(m-1)}(x) + \nu \cdot h_m(x), \quad p = 1 / (1 + \exp(-F_M(x))) \quad (8.2)$$

In equation 8.2 each new tree h_m is fitted to the errors of the current model and added with learning rate ν (Friedman, 2001). Early stopping on 10% of the training rows decides how many trees to keep.

8.4 Model Comparison and Threshold Selection

Table 16: Validation results of the candidate models (n = 135,603)

Model	ROC-AUC	AP	Brier	Log loss	Trees
Dummy (training prevalence)	0.5000	0.0502	0.04771	0.1992	-
Logistic regression	0.5954	0.0743	0.04740	0.1962	-
Logistic regression, balanced weights*	0.5958	0.0740	0.24165	0.6766	-
Random forest (100k rows, earlier model)	0.6458	0.1053	0.04683	0.1921	100
Gradient boosting, lr 0.10, 31 leaves	0.6663	0.1238	0.04624	0.1887	186
Gradient boosting, lr 0.05, 31 leaves	0.6660	0.1235	0.04624	0.1887	284
Gradient boosting, lr 0.03, 63 leaves (selected)	0.6664	0.1241	0.04623	0.1886	366

All three gradient boosting settings beat the random forest by about 0.02 in AUC and 18% in Average Precision. The selected setting has the highest validation AP. Balanced class weights leave the ranking unchanged but make the Brier score five times worse, which is why that model is excluded. An isotonic recalibration was tested by fitting it on one half of the validation set and checking it on the other. It did not improve the Brier score in either half, so it was not applied.

Table 17: Alert-threshold options on the validation set

Rule	Threshold	Flagged	Precision	Recall	Cost*
Maximise F1 (default)	0.0959	6.9%	16.2%	22.2%	34,351

Rule	Threshold	Flagged	Precision	Recall	Cost*
Minimise cost (FP = 1, FN = 5)	0.1569	2.5%	23.2%	11.5%	32,741
Review capacity of 10%	0.0805	10.0%	13.8%	27.4%	36,407

.

The threshold is a business choice rather than a statistical one (Table 17). The F1 rule flags 6.9% of policies. The cost rule, with an illustrative ratio of 5 to 1 between a missed claim and an unnecessary review, flags only 2.5% with higher precision. The capacity rule flags exactly what a review team can handle. The F1 rule is used for the test results below.

8.5 Test Evaluation

Table 18: Test-set results (n = 135,603, scored once)

Metric	Gradient boosting (95% CI)	Earlier random forest	Dummy
ROC-AUC	0.6647 (0.6576 to 0.6708)	0.6394	0.5000
Average Precision	0.1262 (0.1205 to 0.1338)	0.1044	0.0502
Brier score	0.04622 (0.04605 to 0.04637)	0.04687	0.04771
Log loss	0.1888	0.1925	0.1992
Precision at threshold	15.48%	11.81%	n/a
Recall at threshold	21.48%	23.58%	n/a
Policies flagged	6.97%	10.02%	0%
Mean predicted vs observed rate	5.03% vs 5.02%	5.03% vs 5.02%	5.02% vs 5.02%

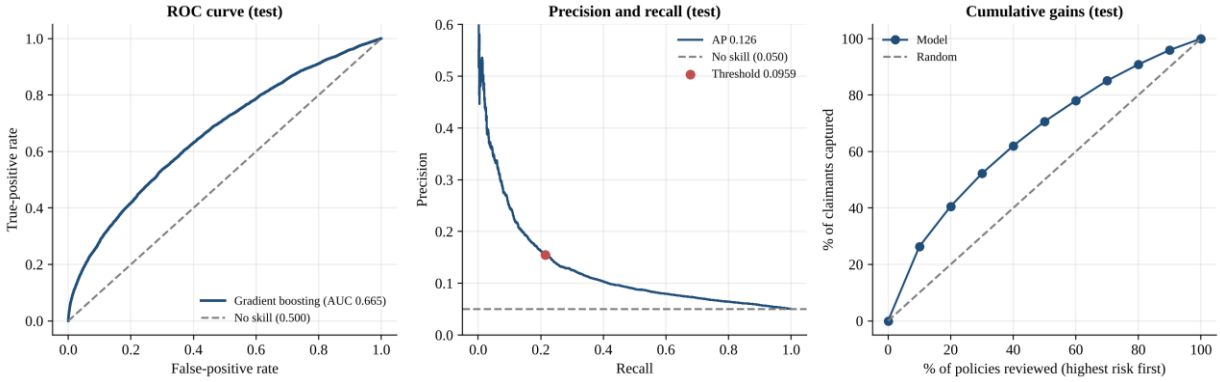


Figure 13: ROC curve, precision and recall, and cumulative gains on the test set

Interpretation: Test results are close to validation results, so selection did not overfit. The earlier random forest's AUC of 0.639 lies outside the new model's 95% interval, so the gain is not due to chance. At the F1 threshold the model flagged 9,450 policies, of which 1,463 claimed. Precision was 15.5%, about three times the base rate, but 5,349 claimants were not flagged (Table 18). The dummy model would score 95% accuracy with no skill at all, which is why accuracy is not reported. The gains curve shows the practical value (Figure 13). Reviewing

the riskiest 10% of policies captures 26.3% of claimants, and the top 1% has a claim rate of 32.0%, 6.4 times the average (Table 19).

Table 19: Claims captured at different review capacities (test)

Top share reviewed	Policies	Claimants	Share of claimants	Precision	Lift
1%	1,356	434	6.4%	32.0%	6.37
5%	6,780	1,200	17.6%	17.7%	3.52
10%	13,560	1,793	26.3%	13.2%	2.63
20%	27,121	2,765	40.6%	10.2%	2.03
30%	40,681	3,557	52.2%	8.7%	1.74

8.6 Calibration, Risk Bands and Stability

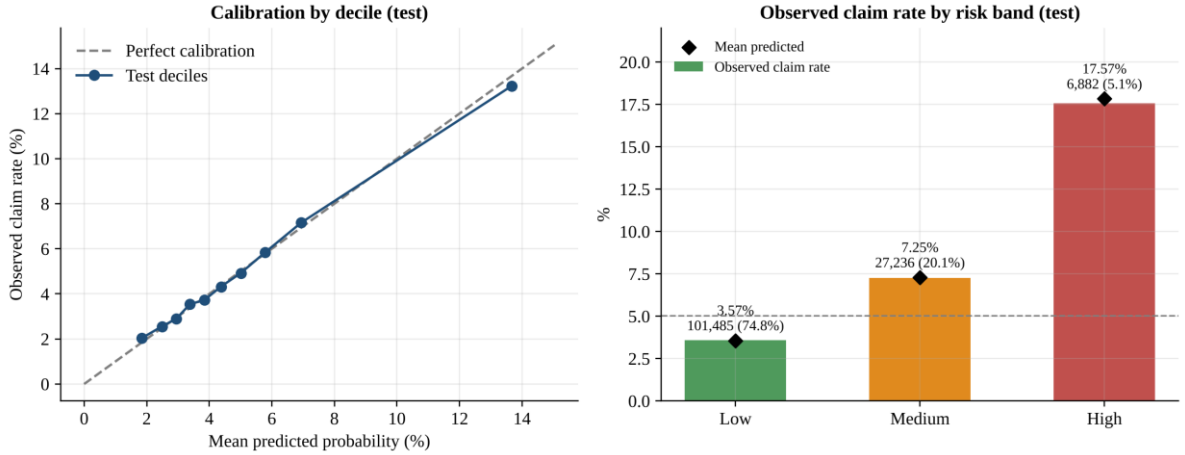


Figure 14: Calibration by decile and observed claim rate by risk band (test)

Interpretation: Predicted and observed rates agree closely in every decile, from 1.84% against 2.04% in the lowest to 13.67% against 13.23% in the highest (Figure 14). The earlier random forest under-predicted its top decile (9.89% against 11.81%). The risk bands are percentiles of validation predictions: Low up to 5.76%, Medium up to 11.24% and High above. On the test set the High band (5.1% of policies) had an observed claim rate of 17.57% against 3.57% for the Low band, and predicted and observed rates agree within each band. Across the six geographic areas, AUC ranged from 0.656 to 0.673 and the ratio of predicted to observed claims from 0.98 to 1.04, so the model behaves consistently across segments.

8.7 Feature Importance and Partial Dependence

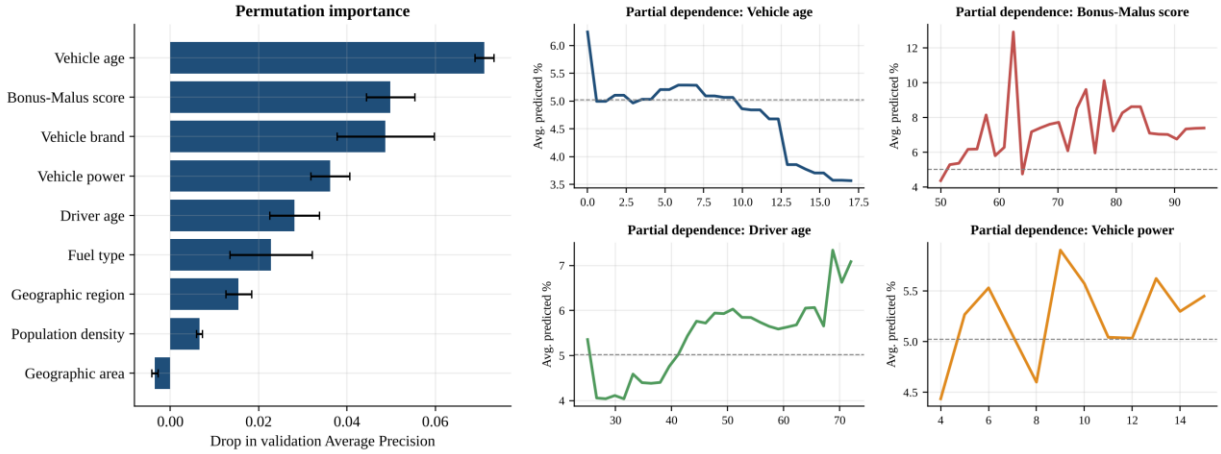


Figure 15: Permutation importance and partial dependence of the selected model

Interpretation: Shuffling vehicle age reduces validation AP by 0.071, the largest effect, followed by Bonus-Malus (0.050), vehicle brand (0.049) and vehicle power (0.036). Geographic area adds nothing once region and density are known. In the partial dependence plots, predicted risk is highest for brand-new vehicles and falls for vehicles older than about 12 years. It rises with Bonus-Malus and is lowest for drivers in their late twenties and thirties (Figure 15). The Bonus-Malus curve is jagged at sparse score values, which suggests the model fits noise there; monotonic constraints would be a sensible refinement. These plots describe the model, not causes.

8.8 Claim-Frequency Model with Exposure

Table 20: Poisson frequency models (claims capped at 4, exposure at 1 year)

Model	Poisson deviance	D^2	AUC of P(claim | exposure)
Constant frequency (validation)	0.6247	0.0%	n/a
Poisson GLM, one-hot (validation)	0.6048	3.2%	0.6295
Poisson gradient boosting (validation, selected)	0.5726	8.4%	0.6835
Selected model on test	0.5731	8.2%	0.6890

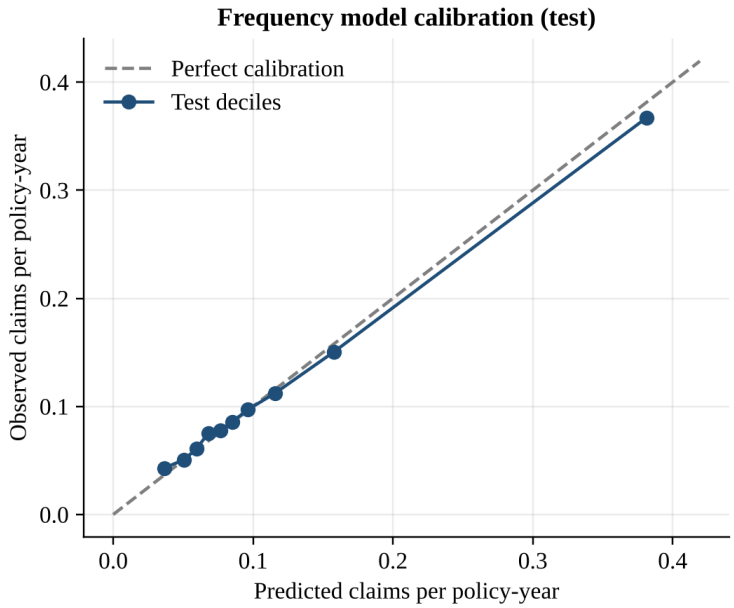


Figure 16: Calibration of the frequency model by decile (test)

Interpretation: The boosted Poisson model explains 8.2% of test deviance, compared with 3.2% for the linear GLM on validation, because claim frequency depends non-linearly on age and Bonus-Malus (Table 20). It predicted 7,221 claims on the test set against 7,225 observed, and its deciles follow the diagonal from 0.037 to 0.381 predicted claims per year (Figure 16). The AUC in the last column uses each policy's own exposure, so it is not comparable with the classification AUC. The frequency model also corrects a weakness of the classifier. Policies that are usually observed for a short time, such as those of young drivers with new cars, can look low-risk to the classifier simply because they had less time to claim (Section 10.3).

8.9 Limitations and Possible Use

The model is trained on French data and has no established validity for ICICI Lombard or Indian motor insurance, where rating factors such as the no-claim bonus differ. A random split cannot show performance drift over time. Bonus-Malus encodes past claims, so part of the skill reflects claim history. No fairness review has been done, and geographic variables could act as proxies for protected characteristics. After local validation, a model of this kind could rank policies for underwriting review or support portfolio segmentation. It should not be used to accept or decline a policy automatically.

9. Anomaly Detection

9.1 Purpose and Method

Anomaly screening looks for policy records that differ from the rest of the portfolio, so that an analyst can check them. Isolation Forest builds random trees that split on random features at random values. Records that are few and different are isolated close to the root, so their average path length E[h(x)] is short (Liu et al., 2008):

$$s(x, n) = 2^{(-E[h(x)] / c(n))} \quad (9.1)$$

Here c(n) is the average path length for n records. The code reports a score that is positive when a record is more isolated than the cut-off implied by a contamination setting of 1%.

9.2 Two Screens

Table 21: Configuration of the two anomaly screens

Setting	Screen A: prospective	Screen B: retrospective
Inputs	Vehicle power, vehicle age, driver age, Bonus-Malus, log density	Screen A inputs plus log exposure, claim count, log claim frequency and claims surprise
When usable	At underwriting (no outcome data)	Only after the period of cover
Claims surprise	Not used	-log10 P(N ≥ observed claims | predicted frequency × exposure)
Forest	200 trees, contamination 1%, fitted on 100,000 records, all records scored	Same
Output per record	Score, flag, top three reasons (robust z-scores), review priority, data-quality rule	Same

9.3 Results

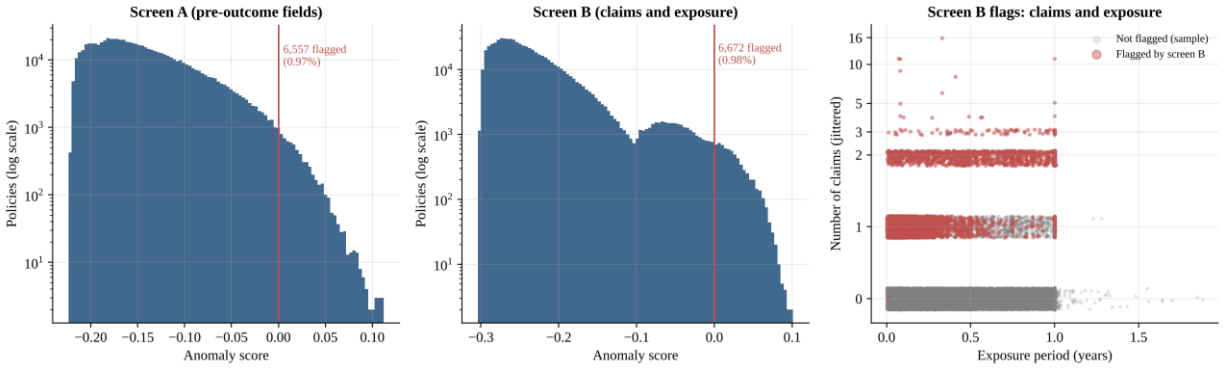


Figure 17: Anomaly scores of both screens and screen B flags by claims and exposure

Table 22: Summary of the two screens

Measure	Screen A	Screen B
Records flagged	6,557 (0.97%)	6,672 (0.98%)
Flagged records with at least one claim	8.1%	99.99%

Measure	Screen A	Screen B
Median Bonus-Malus (flagged / others)	93 / 50	62 / 50
Median vehicle power (flagged / others)	10 / 6	6 / 6
Median exposure in years (flagged / others)	0.41 / 0.49	0.13 / 0.49
Most common first reason	Bonus-Malus (3,393), vehicle power (2,472)	Claim frequency (4,245), claims surprise (1,289)
Mean flag overlap across 5 seeds (Jaccard)	0.69	0.77

.

Interpretation: The two screens find different things (Table 22). Screen A flags unusual profiles, mainly high Bonus-Malus scores combined with powerful or older vehicles, and only 8.1% of these records had a claim. It is a data-quality and profile check that can run before any claim occurs. Screen B flags policies with many claims in a short period (Figure 17). The top record had two claims in 0.0027 years of exposure, with a claims surprise of 7.1, which means a probability below one in ten million given its risk profile. Overlap between random seeds of 0.69 and 0.77 shows the flags are reasonably but not perfectly stable, so borderline cases should not be treated as firm. Separately, simple rules flag 1,328 records (vehicle age above 50, exposure above one year, or more than five claims).

9.4 Interpretation and Leakage

A flag means only that a record is statistically unusual. It may be a data error, a rare but genuine situation or a case worth investigating. The dataset has no fraud labels, claim dates or claim notes, so none of the flagged records can be called fraudulent or even suspicious. Screen B uses claim count and exposure, which are known only after the cover period. It therefore suits retrospective review, and its output must never be used as an input to the prediction models, because that would leak the outcome into the predictors (Kaufman et al., 2012).

9.5 Operational Use

Screen A could run at underwriting to catch data-entry problems. Screen B could feed a periodic review queue sorted by review priority, for example its top 0.5%. In both cases reviewers should record the outcome of each case (data error, legitimate, or referred), so that the precision of the screens can be measured over time. Without that feedback the contamination level remains a workload choice rather than an estimate of how many records are truly anomalous.

10. Prediction Dashboard and Decision Support

10.1 Purpose and Workflow

The last section of the enhanced code launches a Policy Claim Probability Predictor built with ipywidgets inside a Jupyter notebook. It lets an analyst describe a policy and see the model's estimates in context without reading code. It is an educational prototype and has not been deployed or connected to any live system. Figure 18 shows how data move through the framework.

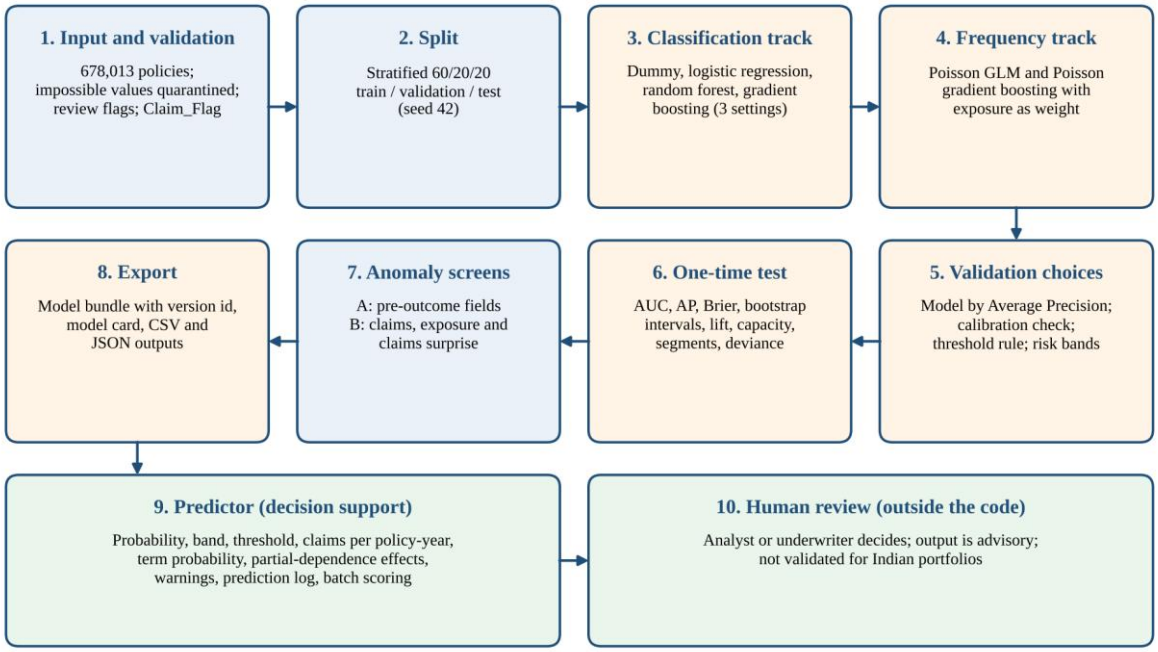


Figure 18: Workflow of the enhanced claim-risk framework

10.2 Implemented Functionality

Table 23: Functions of the predictor and notebook

Function	Implementation	Status
Data loading and validation	Quarantine of impossible values, review flags, quality table	Implemented
Model training and evaluation	Seven candidates, validation table, one-time test, bootstrap intervals, lift and capacity tables	Implemented
Policy inputs	Sliders limited to training ranges, dropdowns of known categories, policy-term slider	Implemented
Outputs	Probability, 0 to 100 score, risk band, alert flag, claims per policy-year, probability for the chosen term	Implemented
Threshold	Slider that changes only the alert label	Implemented
Explanation	Partial-dependence effect of each input value across 300 reference policies	Implemented
Safeguards	Negative values rejected; warnings for out-of-range values, unseen categories and missing values	Implemented

Function	Implementation	Status
Logging and versioning	Each saved prediction appended to a log with a model version id	Implemented
Batch scoring	score_csv() scores a file of new policies	Implemented (code)
Authentication, audit trail, deployment	Not present	Not implemented

10.3 Illustrative Output

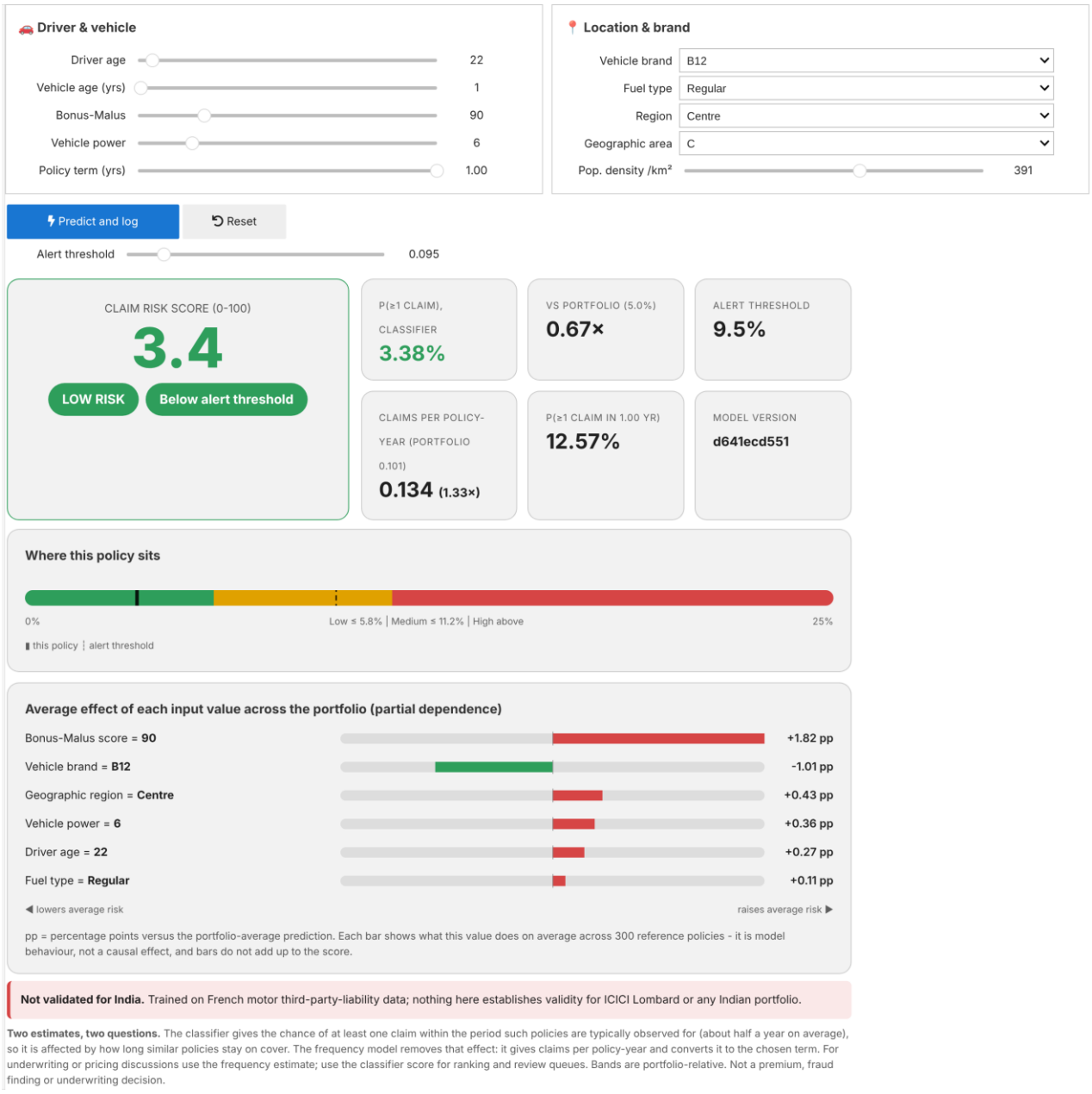


Figure 19: Policy Claim Probability Predictor with its input controls and result panel (driver aged 22, Bonus-Malus 90, one-year-old vehicle)

How to read it: The upper panels hold the input controls: sliders for driver age, vehicle age, Bonus-Malus, vehicle power, policy term and population density, and drop-down lists for brand, fuel type, region and area. Every slider is limited to the range seen in training. 'Predict and log' saves the result with the model version, 'Reset' restores the default policy and the alert-threshold slider moves only the alert label. Any change to an input updates the result panel below at once.

Interpretation: For the default policy (training medians and most frequent categories) the classifier gives 4.09% and the frequency model 0.090 claims per year. For the policy in Figure 19 the classifier gives only 3.38%, but the frequency model gives 0.134 claims per year, 1.33 times the portfolio rate of 0.101, or a 12.6% chance of a claim within one year. The two estimates differ because young drivers with new cars are often observed for short periods, which the classifier cannot separate from low risk. The panel therefore tells the user to rely on the frequency estimate for underwriting discussions and on the classifier score for ranking. The explanation bars show that, on average across the portfolio, a Bonus-Malus of 90 raises predicted risk by 1.82 points and a driver age of 22 by 0.27 points. These averages do not add up to this policy's score.

10.4 Human Oversight and Limitations

An underwriter could use the panel to see where a proposal sits in the portfolio, and a risk manager could use the bands to track changes in the business mix. In every case a person makes the decision, because the model ignores information that the person has and makes frequent errors at the level of a single policy. The panel supports oversight by separating the probability from the threshold, warning about extrapolation and showing a 'Not validated for India' notice. As a prototype it lacks user roles, an audit trail and monitoring. The model bundle is a pickle file, so only trusted files should be loaded.

11. Role of Data Analytics, AI and Generative AI in Insurance

11.1 Types of Analytics

Table 24: Analytics approaches and their use in this project

Approach	What it does	In this project
Descriptive analytics	Summarises what happened with ratios, trends and risk statistics	Implemented (Chapters 6 and 7)
Predictive machine learning	Learns from past outcomes to score new cases	Implemented (Chapter 8)
Anomaly detection	Finds records unlike the majority, without labels	Implemented (Chapter 9)
Generative AI	Produces text or summaries from prompts using large language models	Not implemented; discussed as a future use

Source: Author, drawing on Eling et al. (2022) and OECD (2020).

11.2 Applications and the Place of Generative AI

Across the insurance value chain, analytics supports pricing and underwriting through risk scores, claims management through triage and review queues, risk monitoring through ratio and solvency dashboards, customer service through chatbots and personalised communication, and fraud investigation through anomaly and network analysis (Eling et al., 2022; OECD, 2020). This project used Excel with OpenSolver for the financial and portfolio work, and Python (pandas, scikit-learn, SciPy, Matplotlib and ipywidgets) for the models and the predictor.

Generative AI was not used anywhere in the models or the dashboard. Possible future uses include summarising claim files for adjusters, drafting customer letters for human approval and writing plain-language explanations of a model result. Each carries risks: large language models can state wrong facts fluently (Bommasani et al., 2021), customer data in prompts must follow the Digital Personal Data Protection Act 2023, and generated explanations can sound more certain than the model is. Any such use would need grounding in verified documents, human review and logging (EIOPA, 2021; NIST, 2023).

12. Integrated Discussion and Business Implications

Table 25 sets out each main finding with its method, interpretation, business implication and limitation. The four evidence bases remain separate.

Table 25: Integrated findings

Finding	Method	Interpretation	Implication	Limitation
Combined ratio 102.8% to 103.4%; PAT up 44% in two years	Ratio and trend analysis	Profit comes from investment income	Better risk selection and cost control are the main levers	Three years; 1/n change
Commission ratio 17.0% to 19.0%; expense ratio 29.8% to 32.0%	IRDAI ratios	Costs, not claims, are rising	Monitor acquisition costs by channel	FY2026 ratios rounded
FVCA from +₹680.74 cr to -₹774.47 cr	Balance-sheet analysis	Market risk reaching the balance sheet	Set investment risk limits	Unrealised; may reverse
Sharpe: SBI Life 0.340, ICICI Lombard 0.330, NIFTY 0.291, GIC Re 0.033	Return and risk statistics	Primary insurers rewarded risk better than the reinsurer	Sub-sector choice matters	Window-dependent
Max-Sharpe portfolio 0.372 with no GIC Re	Mean-variance optimisation	Optimiser drops the dominated stock	Use as a diagnostic	Two-stock concentration
Test AUC 0.665, AP 0.126; High band 17.6% vs Low 3.6%	Gradient boosting, one-time test	Useful ranking, weak single-policy prediction	Prioritise reviews after local validation	French data; no time split
Frequency model D^2 8.2%; 7,221 vs 7,225 claims	Poisson model with exposure	Correct basis for annual risk	Use for pricing discussions	French data
Screens flag about 1% each; overlap 346	Isolation Forest	Profile oddities (A) and claim bursts (B)	Review queue with outcome codes	No labels

12.1 Financial Performance, Risk and Investment

ICICI Lombard's profit rose sharply while underwriting stayed marginally loss-making, and about a quarter of FY2025's PBT growth came from lower provisions. Management attention should therefore focus on the combined ratio by segment and on the drift in commission and expenses, rather than on PAT growth alone. The swing in fair value change accounts shows that investment-market risk now matters as much as underwriting risk. In the stock analysis, combining insurers reduced risk, but optimised weights depended heavily on the sample window and on GIC Re's poor early years.

12.2 Predictive Modelling, Anomalies and Governance

The enhanced model ranks risk better than the earlier one and its probabilities are well calibrated, yet at the chosen threshold about five of six flagged policies do not claim and three of four claimants are not flagged. Both facts must be shown to users. The frequency model is the better basis for annual risk because it separates exposure from risk. The anomaly screens produce manageable queues, but their value can only be measured if reviewers record outcomes. The project already applies several good governance practices: leakage controls,

validation-only choices, a single test evaluation, model versioning and persistent disclaimers. Missing are a fairness review, drift monitoring and independent validation (Board of Governors of the Federal Reserve System, 2011). The workbook errors found in Chapter 5 show that spreadsheets need the same review discipline as code.

13. Risk Mitigation and Recommendations

Table 26: Prioritised recommendations

Priority	Area	Recommendation	Evidence	Status
1	Underwriting and cost monitoring	Track the combined ratio by segment on both 1/n and n bases; set triggers for Marine and Fire; monitor commission and expense drift	Sections 6.2, 6.3	Proposed
1	Investment risk	Set limits on fair-value exposure and report VaR and drawdown alongside volatility	Sections 6.4, 7.2	Proposed
1	Data governance	Keep the Validation Log; replace typed values with formulas; record the price source; replace the mid-month June 2026 prices	Section 5.2	Partly done
1	Model validation	Re-train and validate on local, dated data with an out-of-time test before any Indian use	Section 8.9	Proposed
2	Threshold and errors	Set the threshold from real review costs and capacity; never decline automatically; track missed claims	Section 8.4	Option in code
2	Frequency basis	Use claims per policy-year for underwriting discussions and the classifier score for ranking	Sections 8.8, 10.3	Implemented
2	Anomaly review	Route the top 0.5% of screen B to reviewers and record outcomes; run screen A at data entry	Chapter 9	Screens implemented
3	Explainability and oversight	Add monotonic constraints for Bonus-Malus and local explanations such as SHAP; keep human sign-off	Section 8.7	Proposed
3	Monitoring	Track AUC, AP, calibration and population stability monthly with retraining triggers	Section 8.6	Proposed
3	Future data	Collect policy and claim dates, claim notes, investigation outcomes and Indian rating factors	Section 9.4	Proposed

The recommendations follow directly from the findings. Because the combined ratio sits just above 100%, small improvements matter: one percentage point of loss ratio on FY2026 net earned premium of ₹22,264 crore (ICICI Lombard, 2026) is worth about ₹223 crore before tax, roughly 8% of FY2026 PAT. For the models, the most valuable next step is validation on local data with dates. After that comes a threshold based on real costs. Flags should always trigger a review rather than an automatic decision, and standard controls should remain for unflagged policies, since most claimants are not flagged.

14. Limitations and Conclusion

14.1 Limitations

- **Data:** one insurer over three years; price-only stock data with an unrecorded source and a mid-month June 2026 row; French claim data without dates, costs or fraud labels.
- **Methods:** historical means as expected returns and a constant 6% risk-free rate; a random rather than time-based split; anomaly results that cannot be evaluated without labels.
- **Comparability:** the 1/n accounting change affects FY2025 and FY2026; some FY2026 IRDAI ratios are published only as rounded values.
- **Generalisability:** no model result transfers to ICICI Lombard or Indian motor insurance without local re-training and validation.

14.3 Conclusion

The analyses point in the same direction. ICICI Lombard's profitability rests on capital strength and investment income rather than underwriting margin. The stock analysis shows that sub-sector matters for investors in insurance. The claim models show that careful validation can produce a better and more honest risk ranking. The framework is far from a deployable underwriting system, but checking the workbook and re-running the code showed that the results can be reproduced and that their limits can be stated plainly.

References

Board of Governors of the Federal Reserve System. (2011). *Supervisory guidance on model risk management* (SR Letter 11-7). https://www.federalreserve.gov/supervisionreg/srletters/sr1107.htm

Bommasani, R., Hudson, D. A., Adeli, E., Altman, R., Arora, S., von Arx, S., et al. (2021). *On the opportunities and risks of foundation models* (arXiv:2108.07258). https://doi.org/10.48550/arXiv.2108.07258

Breiman, L. (2001). Random forests. *Machine Learning, 45*(1), 5-32. https://doi.org/10.1023/A:1010933404324

Brier, G. W. (1950). Verification of forecasts expressed in terms of probability. *Monthly Weather Review, 78*(1), 1-3.

Chandola, V., Banerjee, A., & Kumar, V. (2009). Anomaly detection: A survey. *ACM Computing Surveys, 41*(3), Article 15. https://doi.org/10.1145/1541880.1541882

Davis, J., & Goadrich, M. (2006). The relationship between precision-recall and ROC curves. In *Proceedings of the 23rd International Conference on Machine Learning* (pp. 233-240). ACM. https://doi.org/10.1145/1143844.1143874

DeMiguel, V., Garlappi, L., & Uppal, R. (2009). Optimal versus naive diversification: How inefficient is the 1/N portfolio strategy? *The Review of Financial Studies, 22*(5), 1915-1953. https://doi.org/10.1093/rfs/hhm075

Denuit, M., Hainaut, D., & Trufin, J. (2019). *Effective statistical learning methods for actuaries I: GLMs and extensions*. Springer.

Dutang, C., & Charpentier, A. (n.d.). *CASdatasets: Insurance datasets* [R package; dataset freMTPL2freq]. https://github.com/dutangc/CASdatasets

Efron, B., & Tibshirani, R. J. (1993). *An introduction to the bootstrap*. Chapman & Hall.

Eling, M., Nuessle, D., & Staubli, J. (2022). The impact of artificial intelligence along the insurance value chain and on the insurability of risks. *The Geneva Papers on Risk and Insurance: Issues and Practice, 47*(2), 205-241. https://doi.org/10.1057/s41288-020-00201-7

European Insurance and Occupational Pensions Authority. (2021). *Artificial intelligence governance principles: Towards ethical and trustworthy artificial intelligence in the European insurance sector*. Publications Office of the European Union.

Fama, E. F., & French, K. R. (2004). The capital asset pricing model: Theory and evidence. *Journal of Economic Perspectives, 18*(3), 25-46. https://doi.org/10.1257/0895330042162430

Fisher, A., Rudin, C., & Dominici, F. (2019). All models are wrong, but many are useful: Learning a variable's importance by studying an entire class of prediction models simultaneously. *Journal of Machine Learning Research, 20*(177), 1-81.

Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *The Annals of Statistics, 29*(5), 1189-1232. https://doi.org/10.1214/aos/1013203451

Hosmer, D. W., Jr., Lemeshow, S., & Sturdivant, R. X. (2013). *Applied logistic regression* (3rd ed.). Wiley.

ICICI Lombard General Insurance Company Limited. (2024). *Integrated annual report 2023-24*.

ICICI Lombard General Insurance Company Limited. (2025). *Integrated annual report 2024-25*.

ICICI Lombard General Insurance Company Limited. (2026). *Integrated annual report 2025-26*.

Kaufman, S., Rosset, S., Perlich, C., & Stitelman, O. (2012). Leakage in data mining: Formulation, detection, and avoidance. *ACM Transactions on Knowledge Discovery from Data, 6*(4), Article 15. https://doi.org/10.1145/2382577.2382579

Lemaire, J. (1995). *Bonus-malus systems in automobile insurance*. Kluwer Academic Publishers.

Lintner, J. (1965). The valuation of risk assets and the selection of risky investments in stock portfolios and capital budgets. *The Review of Economics and Statistics, 47*(1), 13-37. https://doi.org/10.2307/1924119

Liu, F. T., Ting, K. M., & Zhou, Z.-H. (2008). Isolation forest. In *2008 Eighth IEEE International Conference on Data Mining* (pp. 413-422). IEEE. https://doi.org/10.1109/ICDM.2008.17

Markowitz, H. (1952). Portfolio selection. *The Journal of Finance, 7*(1), 77-91. https://doi.org/10.1111/j.1540-6261.1952.tb01525.x

Michaud, R. O. (1989). The Markowitz optimization enigma: Is 'optimized' optimal? *Financial Analysts Journal, 45*(1), 31-42. https://doi.org/10.2469/faj.v45.n1.31

National Institute of Standards and Technology. (2023). *Artificial intelligence risk management framework (AI RMF 1.0)* (NIST AI 100-1). https://doi.org/10.6028/NIST.AI.100-1

Noll, A., Salzmann, R., & Wüthrich, M. V. (2018). *Case study: French motor third-party liability claims*. SSRN. https://doi.org/10.2139/ssrn.3164764

OECD. (2020). *The impact of big data and artificial intelligence (AI) in the insurance sector*. OECD Publishing.

Ohlsson, E., & Johansson, B. (2010). *Non-life insurance pricing with generalized linear models*. Springer.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research, 12*, 2825-2830.

Penman, S. H. (2013). *Financial statement analysis and security valuation* (5th ed.). McGraw-Hill Education.

Phua, C., Lee, V., Smith, K., & Gayler, R. (2010). *A comprehensive survey of data mining-based fraud detection research* (arXiv:1009.6119). https://doi.org/10.48550/arXiv.1009.6119

Piotroski, J. D. (2000). Value investing: The use of historical financial statement information to separate winners from losers. *Journal of Accounting Research, 38*(Suppl.), 1-41. https://doi.org/10.2307/2672906

Power, D. J. (2002). *Decision support systems: Concepts and resources for managers*. Quorum Books.

Rejda, G. E., McNamara, M. J., & Rabel, W. H. (2020). *Principles of risk management and insurance* (14th ed.). Pearson.

Saito, T., & Rehmsmeier, M. (2015). The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE, 10*(3), e0118432. https://doi.org/10.1371/journal.pone.0118432

Sharpe, W. F. (1964). Capital asset prices: A theory of market equilibrium under conditions of risk. *The Journal of Finance, 19*(3), 425-442. https://doi.org/10.1111/j.1540-6261.1964.tb02865.x

Sharpe, W. F. (1966). Mutual fund performance. *The Journal of Business, 39*(1), 119-138. https://doi.org/10.1086/294846

Shmueli, G., & Koppius, O. R. (2011). Predictive analytics in information systems research. *MIS Quarterly, 35*(3), 553-572. https://doi.org/10.2307/23042796

Treynor, J. L. (1965). How to rate management of investment funds. *Harvard Business Review, 43*(1), 63-75.

Viaene, S., Derrig, R. A., Baesens, B., & Dedene, G. (2002). A comparison of state-of-the-art classification techniques for expert automobile insurance claim fraud detection. *Journal of Risk and Insurance, 69*(3), 373-421. https://doi.org/10.1111/1539-6975.00023

Wüthrich, M. V., & Merz, M. (2023). *Statistical foundations of actuarial learning and its applications*. Springer. https://doi.org/10.1007/978-3-031-12409-9

Zadrozny, B., & Elkan, C. (2002). Transforming classifier scores into accurate multiclass probability estimates. In *Proceedings of the Eighth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 694-699). ACM. https://doi.org/10.1145/775047.775151

Appendix A: Supplementary Model Tables

Table 27: Lift by decile of predicted risk (test)

Decile	Policies	Claims	Mean predicted	Claim rate	Lift	Cumulative claims
1 (highest)	13,561	1,794	13.67%	13.23%	2.63	26.3%
2	13,560	971	6.94%	7.16%	1.43	40.6%
3	13,560	792	5.79%	5.84%	1.16	52.2%
4	13,561	668	5.01%	4.93%	0.98	62.0%
5	13,560	586	4.37%	4.32%	0.86	70.6%
6	13,560	505	3.84%	3.72%	0.74	78.0%
7	13,561	481	3.38%	3.55%	0.71	85.1%
8	13,560	392	2.94%	2.89%	0.58	90.9%
9	13,560	347	2.49%	2.56%	0.51	96.0%
10 (lowest)	13,560	276	1.84%	2.04%	0.41	100.0%

Table 28: Model stability by geographic area (test)

Area	Policies	Observed rate	Mean predicted	Predicted / observed	AUC
A	20,750	4.47%	4.66%	1.04	0.656
B	15,171	4.73%	4.85%	1.03	0.658
C	38,278	4.96%	4.85%	0.98	0.668
D	30,402	5.25%	5.21%	0.99	0.673
E	27,379	5.35%	5.34%	1.00	0.657
F	3,623	5.74%	5.88%	1.02	0.672

Appendix B: Reproducibility Notes

Table 29: Files, settings and software

Item	Detail
Workbook	CIA -4 FRA (validated).xlsx, with a Validation Log sheet of 36 checks and corrections
Visualisation code	CIA-4 portfolio visualisation script and FRA_additional_visualizations.py; charts redrawn from the corrected workbook
Claim-risk code	Insurance_Claim_Risk_Analytics_ENHANCED.py (single notebook cell)
Dataset	freMTPL2freq 78.csv: 678,013 rows, 12 columns
Settings	Seed 42; stratified 60/20/20 split; gradient boosting with early stopping; 200 bootstrap samples; Isolation Forest with 200 trees, contamination 1%, 100,000 fitting rows
Software	Python 3.13, scikit-learn 1.9.1, pandas 3.0, NumPy 2.5, SciPy 1.18, Matplotlib 3.11, ipywidgets 8.1
Reproducibility	Two complete runs of the enhanced code gave identical metrics, thresholds and flags
Earlier model	Random forest results come from Insurance_Claim_Risk_Analytics_FIXED.py on the same split, reproduced exactly

.